



KENYATTA UNIVERSITY

**INSTITUTE OF OPEN DISTANCE & e-LEARNING
IN COLLABORATION WITH
SCHOOL OF ECONOMICS DEPARTMENT:
ECONOMETRICS AND STATISTICS**

**UNIT CODE & NAME:
EES 100 – MATHEMATICS FOR ECONOMISTS I**

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KENYATTA UNIVERSITY

SCHOOL OF ECONOMICS

EES 100: Mathematics for Economists I**COURSE OUTLINE****1. INTRODUCTION**

- ✓ Need for Mathematics in Economic Analysis

2. SET THEORY

- ✓ Introduction- Definitions
- ✓ Methods of set representation
- ✓ The Venn Diagram

3. FUNDAMENTAL TECHNIQUES IN ALGEBRA

- ✓ Rules of Algebraic Operations
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- ✓ Rules of Differentiation
- ✓ Economic Application of derivatives

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8. MAXIMA AND MINIMA

- ✓ First and second derivatives
- ✓ Tests for relative maximum
- ✓ Tests for relative minimum
- ✓ Economic Applications of maxima and minima

9. PARTIAL DIFFERENTIATION

- ✓ Rules for partial differentiation
- ✓ Economic Application of partial derivatives

READING LIST

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Jacques, I. (1999), *Mathematics for Economics and Business*, 3rd ed. Addison Wesley Longman.

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CHAPTER 1



INTRODUCTION

1.0



1.1 Need for Mathematics in Economic Analysis

Mathematics is an invaluable tool at all levels of the study of economics, ranging from the statistical expression of real world trends to the development of fully abstract economic systems

1. Mathematics provides foundations for empirical propositions about economic variables. Proposition like “a 10 percent increase in the price of a good causes a 5 percent drop in the quantity demanded of the good” is a mathematical expression of demand function.
2. Statistics which is a branch of mathematics is used by economists to transform raw data from the real world into numerical generalizations. Such generalizations are used by economists to construct a network of interlocking relationships which enable them to draw conclusions about economic variables that are related to each other only directly.
3. Economists construct mathematical representations (models) of markets and communities to understand better how they work. This is done by picking out the most important aspects of a situation and then expressing them mathematically.

Thus the mathematical model reduces the complexity of the real world to manageable proportions

4. Mathematics is also used to actively generate and explore new theoretical ideas. For example, economists use mathematical techniques such as logical deductions to derive theorems which apply to a wide variety of economic situations.

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5. Mathematics is not only a powerful tool for gaining insights from models of the economy, but is also needed to broaden applicability of a model that is too narrowly constructed to be useful. for example, the simple models for production or sale of two goods, can be extended so as to handle more information at one time

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CHAPTER 2



2.0 SET THEORY

2.1 Definition of a set



A **set** is a collection of distinct objects. For example, we can consider all the oceans in the world to be a set.

If A is a set; $A = \{4,6,8,13\}$, the objects in the set, the integers 4,6,8,and 13 are referred to as the **members** or **elements** of the set. The elements can be listed in any order.

The notation $\in$ is used to indicate membership of a set. For example, in the set A we can write $4 \in A$ or $4 \in \{4,6,8,13\}$

A set S is said to be a **subset** of another set T if every element in S is a member of T. a subset is denoted by $\subset$. If $B = \{4,8\}$ then since the integers 4 and 8 are members of set A

we can write $B \subset A$ or $\{4,8\} \subset \{4,6,8,13\}$

Other definitions

1. Finite and infinite set

A set can be classified as finite set or infinite set depending on the number of elements that it has. a finite set has a finite number of elements while an infinite set has an infinite number of elements. If P and S are sets: $P = \{2,4,6,\dots,20\}$; $S = \{1,3,5, \dots, \dots, \dots\}$

P is a finite set while S is an infinite set.

2. Universal set

A universal set refers to a set that contains all the elements that an analyst wishes to

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study. if a researcher wishes to study the fresh water lakes in a country X then the

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universal set will be all the fresh water lakes in that country. The various classifications of the fresh water lakes in the country would be sub sets of the universal set

The notation U is generally used to denote universal sets. Given a universal set we can derive its subsets. for example if $U = \{1,2,3\}$ the subsets are $\{ \}$, $\{1\}$, $\{2\}$, $\{3\}$, $\{1,2\}$, $\{1,3\}$, $\{2,3\}$ and $\{1,2,3\}$.

This means that both the Null set, $\{ \}$, and the Universal set, U , are subsets of U

3. The Null set

It is a set that contains no element hence is also referred to as the empty set. It is designated by a Greek letter $\emptyset$ or $\{ \}$.

Note that the sets $\{ \}$ and $\{0\}$ are not the same. The former has no element in it while the latter has one element in it.

4. Equal sets

Two sets A and B are said to be equal if every member of a set A belongs to B and every element of set B belongs to A . That is the two sets contain the same elements. For example if $A = \{a, d, c, b\}$ and $B = \{d, c, a, b\}$ are equal sets.

2.2 Methods of Set Representation



Capital letters are used to represent sets.

There are two different methods of representing members of a set.

- i) Descriptive method
- ii) Enumerative method

In the *descriptive method*, members of a set are presented in such a way that one can determine the elements of the set without difficulty. For example, P

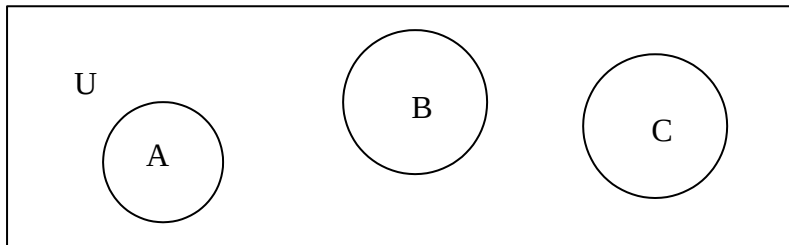
$$= \{x \mid x = 0, 1, 2, \dots, 7\} \text{ or } P = \{x \mid 0 \leq x \leq 7\}$$

The *enumerative method* requires that one writes all the members of the set within the curly brackets. For example the set P could be written as follows $P = \{0,1,2,3,4,5,6,7\}$

2.3 The Venn diagram



The Venn diagram provides a simple way of representing sets and relations between sets. It consists of a rectangle that represents the universal set. Subsets of the universal set are represented by circles drawn within the rectangle, or the universe. Consider the Venn diagram below;



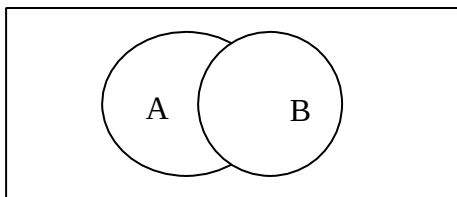
In the Venn diagram above, U is the universal set, while A, B and C are subsets of U.

2.4 Set Operations



i) *The intersection of sets*

Intersection of sets A and B, is a subset of U containing elements which belong to A and B. It is denoted by the symbol $A \cap B$. It can also be represented diagrammatically as below. The shaded area represents the intersection of A and B



Numerical example 1

Given the universal set T and its subsets A and B

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$$T =$$

$$\{1, 2, 3, 4, 5, 6, 7, 8\} \quad A$$

$$= \{3, 4, 5, 6\}$$

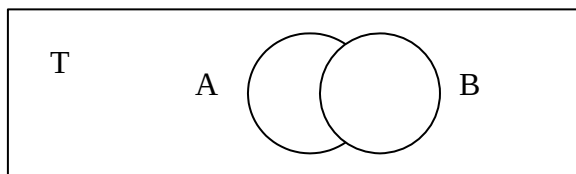
$$B = \{1, 3, 4, 7, 8\}$$

Determine the intersection of A and B

Solution

The intersection of A and B is a Subset of T, containing elements that belong to both A and B hence

$$A \cap B = \{3, 4, 5, 6\} \cap \{1, 3, 4, 7, 8\} = \{3, 4\}$$



Numerical example 2

Consider the following universal set T and its subsets C, D and E:

$$T = \{0, 2, 4, 6, 8, 10, 12\}$$

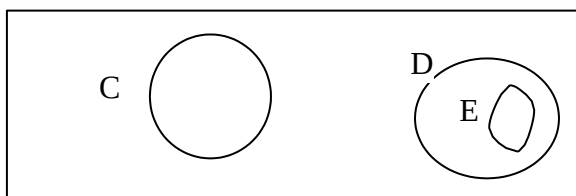
$$C = \{4, 8\}$$

$$D = \{10, 2, 0\} \quad E = \{0\}$$

Find $C \cap D \cap E$

$$C \cap D \cap E = \{4, 8\} \cap \{10, 2, 0\} \cap \{0\} = \{ \} = \emptyset$$

This means that there is no element that belongs to all the subsets



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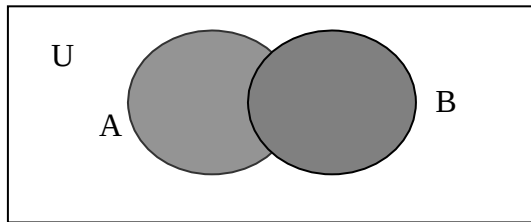
Sets that have no elements in common are said to be *mutually exclusive* or *disjointed*. Sets C and D in the above example are disjointed.

ii) The union of sets

The union of two sets A and B is the set of elements of U which are members of either A or B. it is denoted by the symbol $\cup$. In the figure below, the shaded area represents $A \cup B$

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Numerical example

Consider the universal set T and its subsets A, B and C below

$$T = \{a, b, c, d, e, f\}$$

$$A = \{a, d\}$$

$$B = \{b, c, f\} \quad C = \{a, c, e, f\}$$

Find $A \cup B$, $A \cup C$, $B \cup C$, $A \cup B \cup C$

Solution

$$\text{i) } A \cup B = \{a, d\} \cup \{b, c, f\} = \{a, b, c, d, f\}$$

$$\text{ii) } A \cup C = \{a, d\} \cup \{a, c, e, f\} = \{a, c, d, e, f\}$$

$$\text{iii) } B \cup C = \{b, c, f\} \cup \{a, c, e, f\} = \{a, b, c, e, f\}$$

$$\text{iv) } A \cup B \cup C = \{a, d\} \cup \{b, c, f\} \cup \{a, c, e, f\} = \{a, b, c, d, e, f\} = T$$

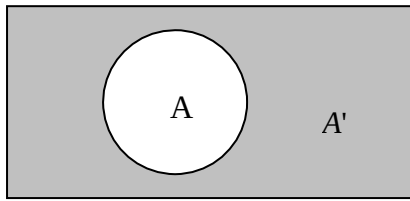
iii) The complement of a set

The complement of a set A is a subset of U containing elements that do not belong to A.

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it is denoted by A' , represented by the shaded area in the figure below



Example

For universal set $T = \{1,2,3,4,5\}$ and its subsets $A = \{2,3\}$ and $B = \{5\}$

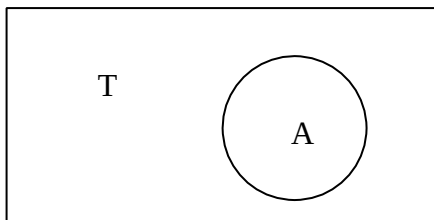
$$A' = \{2,3\}' = \{1,4,5\}$$

$$B' = \{5\}' = \{1,2,3,4\}$$

Laws of Set algebra



Given the Venn diagram



Where T is a Universal set and A is its subset, the following laws can be deduced

$$A \cap \emptyset = A$$

$$A \cap T = T$$

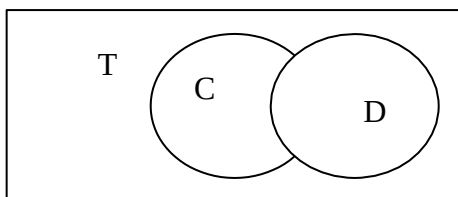
$$A \cap A = A$$

$$A \cap A = A$$

$$A \cap T = A$$

$$A \cap A' = T$$

$$A \cap A' = \emptyset$$



From the Venn diagram two additional laws can be deduced that

$$C \cap D = D \cap C$$

$$C \cap D = D \cap C$$

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CHAPTER 3



3.0

FUNDEMENTAL TECHNIQUES IN ALGEBRA



3.1 Rules of Algebraic Operations



3.1.1 Multiplication rules



1. If two numbers have like signs, their product is positive. For example;

$$(+ \square)(+ \square) = +\square\square$$

$$(- \square)(- \square) = +\square\square$$

2. If two numbers have unlike signs, their product is negative. For example

$$(+ \square)(- \square) = -\square\square$$

$$(- \square)(+ \square) = -\square\square$$

3.1.2 Quotient rules



1. If two numbers have like signs, their quotient is positive. For example;

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$$\begin{array}{r} +\square \quad \square + \square \quad \square \\ = \end{array}$$

$$\begin{array}{r} -\square \quad \square \\ = \\ -\square \quad \square \end{array}$$

2. If two numbers have unlike signs, their quotient is negative. For example

$$\begin{array}{r} +\square = -\square \\ -\square \quad \square \end{array}$$

$$\begin{array}{r} -\square = -\square \\ +\square \quad \square \end{array}$$

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3.1.3 Rules for combined operations



Many problems involve several operations. Sometimes there are brackets used for ordering the operations. For example;

$$3 \times 2 - 4 \div 1 - 2(2 - 4) + 2 \times 3$$

The following rules will be useful in solving such problems.

1. If brackets are present in a problem, first compute the expression within the brackets in the following order
 - First deal with multiplications and divisions - Then deal with additions and subtractions.

Numerical example

Evaluate $(4 \times 2 + 16) \div 2 - 6 + 3 \times 4 + 18 \div 6 - (5 - 2)$

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Solution

Starting with the multiplication and division operations, then to additions and subtraction

$$\begin{aligned} & (4 \times 2 + 16) \times 2 - 6 + 3 \times 4 + 18 \times 6 - (5 - 2) \\ & = (8 + 16) \times 2 - 6 + 12 + 3 - (5 - 2) \\ & = 24 \times 2 - 6 + 12 + 3 - 3 \\ & = 48 - 6 + 12 + 3 - 3 \\ & = 54 \end{aligned}$$

2. If brackets are absent in a problem, compute the problem in the following order
- First deal with multiplications and divisions
 - Then deal with additions and subtractions

Example

Simplify the following

$$12 \times 2 - 6 + 3 \times 6 \times 9 + 30 \times 15$$

Solution

$$\begin{aligned} & 12 \times 2 - 6 + 3 \times 6 \times 9 + 30 \times 15 \\ & = 24 - 6 + 2 + 2 \\ & = 22 \end{aligned}$$

3.2 EXPONENTIALS AND LOGARITHMS



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3.2.1 Meaning of Exponents



The product of six factors, each of which is X can be written as

$X \cdot X \cdot X \cdot X \cdot X \cdot X = X^6$ In the expression, X^6 which is the sixth power of X, X is the **base** and 6 is the **exponent**.

Laws of Exponents

1. Multiplication Law

$$X^a \cdot X^b = X^{a+b}$$

For example; $X^2 \cdot X^3 = X^{2+3} = X^5$

2. Quotient Law

$$\frac{X^a}{X^b} = X^{a-b}$$

For example; $\frac{X^5}{X^3} = X^{5-3} = X^2$

3. Zero Power Law

$$X^0 = 1$$

This can be illustrated by use of the quotient rule

For example $\frac{X^2}{X^2} = \frac{X \cdot X}{X \cdot X} = 1 = X^{2-2} = X^0$

4. The Power Law

$$(X^a)^b = X^{ab}$$

For example; $(X^2)^3 = X^2 . X^2 . X^2 = X^{2+2+2} = X^6$

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5. Reciprocal Law

$$\frac{1}{X^a} = X^{-a} ; \frac{1}{X^{-a}} = X^a$$

Example; $X^2 = X^2 . X^{-5} = X^{2-5} = X^{-3} = \frac{1}{X^3}$

—
 X^5

X^3

3.2.2 Meaning of Logarithms



In the exponential expression $y = a^x$, where **a** is the base and **x** the exponent, we say that the logarithm of y to the base of a is x. that is to say that *the logarithm of y to the base a is the power to which a must be raised so as to obtain y*

Examples;

$$\log_{10} 100 = \log_{10} 10^2 = 2$$

$$\log_3 27 = \log_3 3^3 = 3$$

$$\log_5 5 = \log_5 5^1 = 1$$

Laws of logarithmic operations

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1. Multiplication law

$$\log_a(x.y) = \log_a x + \log_a y$$

2. Quotient law

$$\log \frac{x}{y} = \log_a x - \log_a y$$

3. Power law

$$\log_a(x^a) = a \log x$$

4. Logarithm of a number to its base

$$\log_a a = 1$$



LINEAR FUNCTIONS

4.0



4.1 The Concept of a Function



A function is a mathematical relationship in which the values of a single dependent variable are determined from the values of one or more independent variables. That is, for every value assigned to the independent variable, a unique value of the dependent variable is determined. For example in the expression;

$$y = a + bx$$

both **x** and **y** are variables. This is because they may assume different values throughout the analysis. However, **a** and **b** are constants because they are fixed numbers.

The variable **y** is a dependent variable. It is dependent in that its values are generated from values of **x**. In this case **x** is an independent variable.

The collection of all the values of the independent variable for which the function is defined is referred to as the *domain* of the function, while the collection of all the values of the dependent variable defined by the function.

The functional relationship between variables can also be expressed generally as $y = f(x)$

The customary symbols used for denoting functions include *f*, *F*, *g*, *G*. in cases where two variables say **y** and **z** are different functions of an independent variable, **x**, then after denoting the functional relation between **y** and **x** as $y = f(x)$, a different notation should be used to represent the relation between **z** and **x**.

4.2 Functions of a Single Independent Variable



In the function $y = a + bx$, each value of x uniquely determines a single y value. That is, corresponding to each value of x , there exists one and only one value of y . This gives rise to a set of ordered pairs. In addition, the functional relationship gives us a rule by which the set X can be mapped into a set Y . Such a rule enables us to map each point of the domain (collection of x -values) to be found on one line segment into the corresponding points on the range (corresponding collection of y values) to be found on another line segment. If we place the domain on the horizontal axis and the range on the vertical axis, we obtain a two dimensional figure, where the association between x and the corresponding y value is shown by a set of ordered pairs such as

(x_1, y_1) , (x_2, y_2) and (x_3, y_3)

4.3 Graphical Presentation

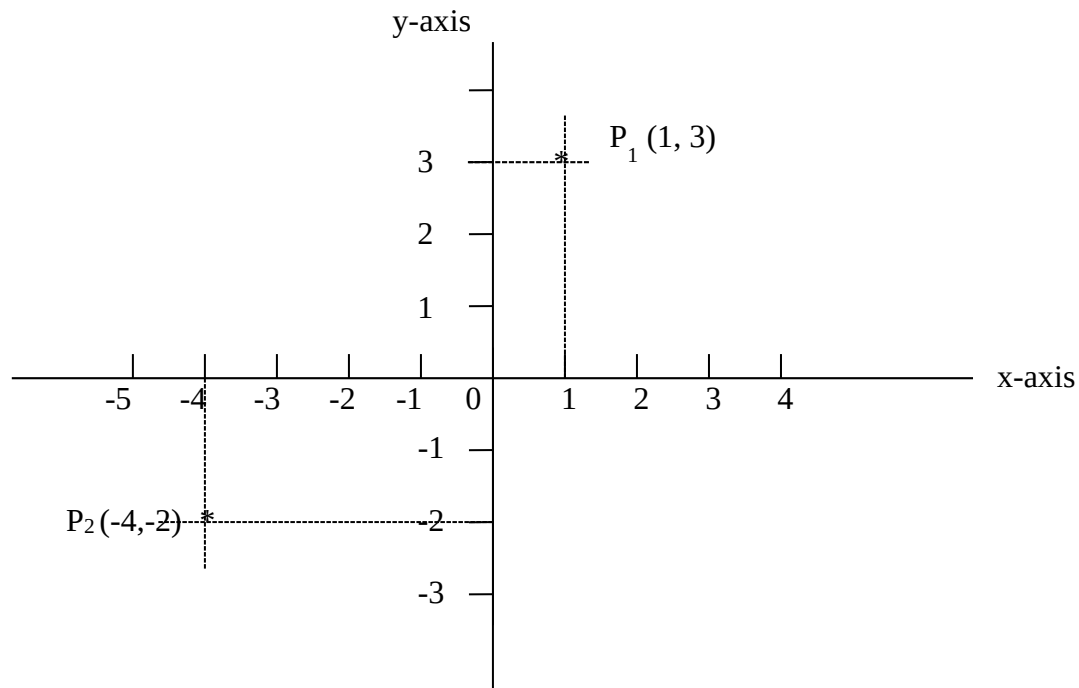


Graphical presentation of functions is important in economic analysis because it brings out more clearly the characteristics of the function in consideration. The graphs mostly used in economic analysis are two dimensional. To graph a two dimensional curve, two directions of axes conventionally referred to as vertical axis (y -axis) and horizontal axis (x -axis) are used.

The two axes cross at the origin. From the origin, the positive and negative values of y and x are measured. All points to the right of the y -axis give positive values of x while those to the left give negative values. Similarly, all points above the x -axis constitute positive values of y while those below constitute negative values.

Points are described by an ordered pair of real numbers in brackets/parenthesis, (x_i, y_i) , referred to as coordinates. The first numbers always refer to the x -coordinate while the second refers to the y -coordinate. For example, for the points P_1 and P_2 in the diagrams below the numbers 1 and -4 are x -coordinates while the numbers 3 and -2 are y -

coordinates.



Since the graph of a function $y = f(x)$ is defined as the totality of all points whose coordinates satisfy the functional relationship. Therefore to draw the graph of a function, we need to obtain a set of ordered pairs (x, y) from the function by;

- Assigning arbitrary values for the independent variable, x , and then
- Calculating the corresponding values of the dependent variable.

The points are then plotted on a coordinate plane.

Example

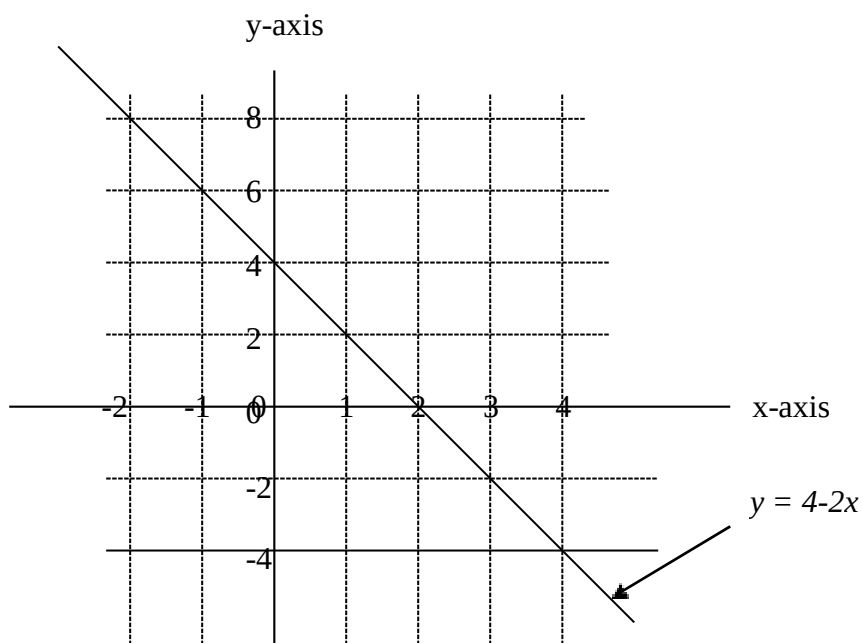
Draw the graph for the function $y = 4 - 2x$

Solution

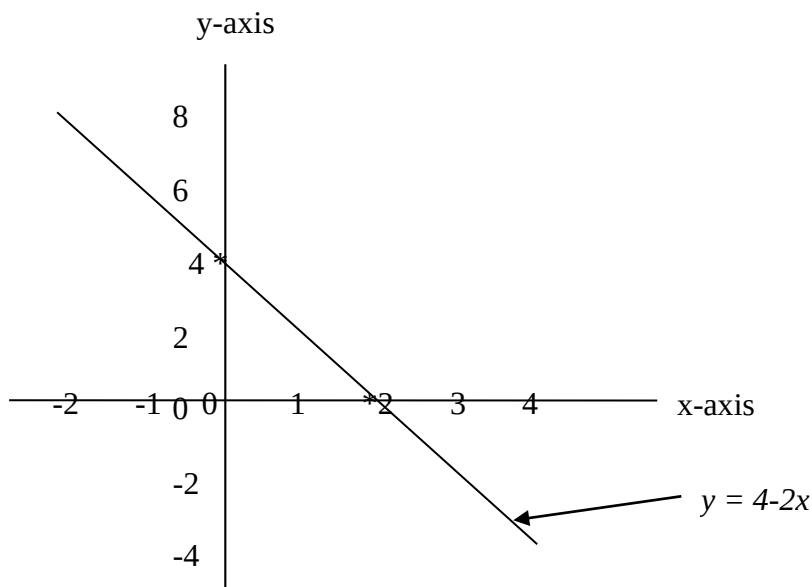
Assuming arbitrary values for x and calculating corresponding values of y for the function yields the following ordered pairs.

x	-2	-1	0	1	2	3	4
y	8	9	4	2	0	-2	-4

Plotting these ordered pairs of (x, y) on the coordinate plane and joining yields the following graph



Note that the graph of a linear function is a straight line. The simplest way to draw a straight line is to determine two ordered pairs of (x,y) that satisfy the function then draw the line. For example for the above example we could have just used two points, e.g. $(0,4)$ and $(2,0)$ to draw the graph thus;



4.4 The Slope of Linear Functions



Given two linear functions;

$$y = 4 - 2x \text{ and } y = 3x - 6$$

These functions can be written generally in the form $y = c + mx$. Therefore

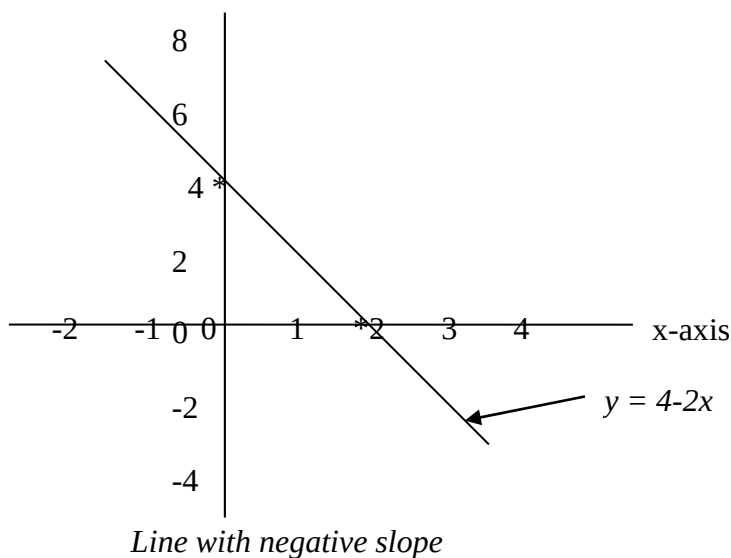
$$y = 4 - 2x \text{(1)}$$

$$y = -6 + 3x \text{(2)}$$

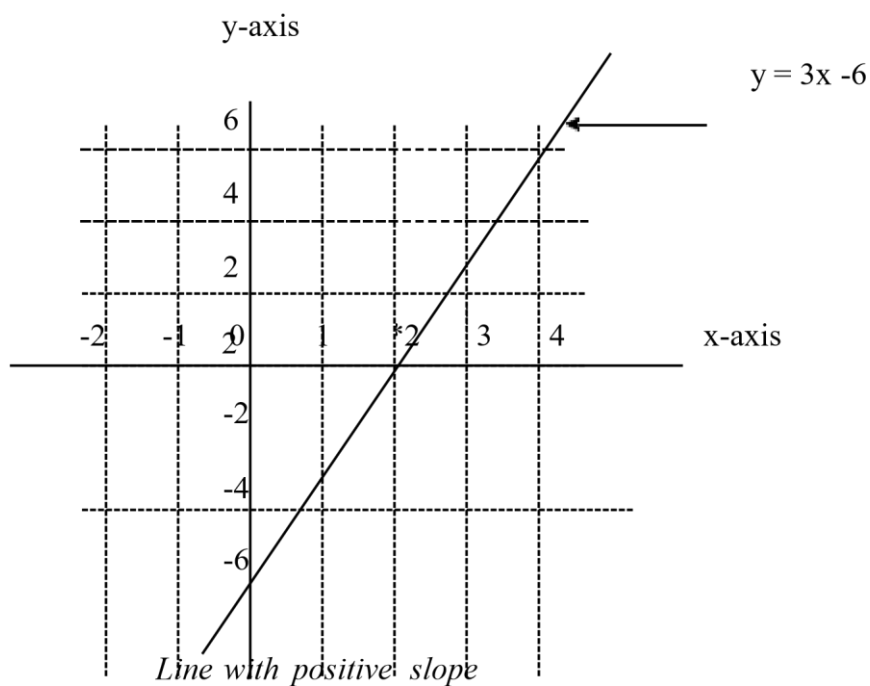
The constant term in each of the two functions gives us the y-intercept (the value of y when $x = 0$).

The coefficient of the independent variable in the linear function represents the slope of the function. A positive coefficient gives a positive slope and a negative coefficient will give a negative slope. The graphs of the two functions confirm this.

y-axis



The slope of the function $y = 4 - 2x$ is -2. A linear function with a negative slope will therefore have a graph sloping downwards.

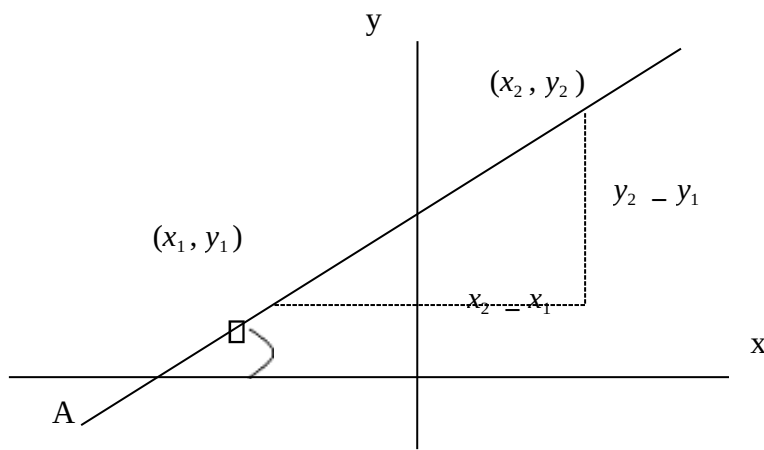


The slope of the function $y = 3x - 6$ is positive. A linear function with a positive slope will therefore slope upwards. *How to compute the slope of a line*

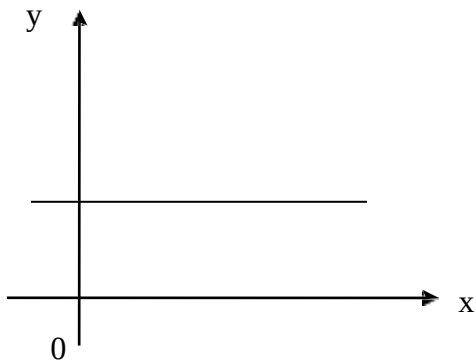
The slope of a line AB, whose two coordinates are given by (x_1, y_1) and (x_2, y_2) is given by

Slope of AB = $\frac{\text{Change in } y}{\text{Change in } x} = \frac{y_2 - y_1}{x_2 - x_1} = \tan \theta$; Where θ is the angle of the line's inclination.

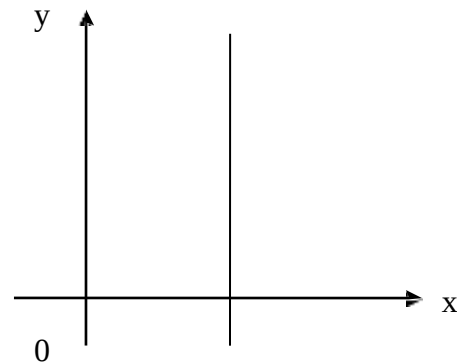
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Apart from negative and positive slopes above, slopes can also be zero or undefined. A horizontal line, i.e. a line parallel to the x-axis has a zero slope. On the other hand the slope of a vertical line, i.e. a line parallel to the y-axis, is undefined.



a) zero slope



b) undefined slope

4.5 Functions and Models in Economic Analysis



Types of functions in economic analysis

Linear functions arise in different ways in economic analysis, and they can be categorized on the basis of the roles they play. On the basis of this criterion, two types of functions can be identified; **behavioral** and **technical** functions.

Behavioral functions specify the manner in which different groups in an economy will behave in response to changes in various explanatory factors or variables. Good examples of such functions are the consumption function, and investment function. Consumption

function describes how the consumers expenditures on consumption goods and services, C , are related to disposable income, Y^d (or income after tax). it is expressed as follows

$$C = a + bY^d \quad a, b \geq 0 \text{ and } Y^d = Y - T$$

where Y = total income or income before tax, T = total tax' and a, b are constants.

Investment function describes the relationship between the firm's investment expenditures, I , and the rate of interest, r , and is expressed as $I = I_0 - I_1r$; $I_0, I_1 \geq 0$

A technical function will be defined here as a function which describes technological or institutional relationships e.g. a production function, which describes the relationship between various inputs and the output.

Models in economic analysis

Our main task in economic theory is an attempt to representation of the actual economy. Since the actual economy is very complex, the most logical thing to do is to

- (i) Take a particular problem/ area you wish to analyze in the economy
- (ii) Single out what you would consider to be some of the major factors relevant to the problem under study
- (iii) Find out the interrelationship among these factors. Such a simplified analytical framework or structure is called an economic model.

An economic model can therefore be defined as a simplified analytical framework or structure representing the relationship among factors or variables in the economy.

We can express such analytical structures in a verbal or mathematical form. The consumption function, the investment function and the tax function are all examples of economic models.

There are two types of variables in economic models; the **endogenous** and the **exogenous** variables. The endogenous variables are the variables that the model is intended to explain, and their values are determined within the model by solving the equation(s) of the model. These variables constitute the unknowns of the model. The exogenous variables on the other hand are not determined within the model. Their magnitudes are obtained from outside the model. That is, their magnitudes are taken as given to the analyst, hence they constitute the knowns of the model.

Economic applications of linear functions

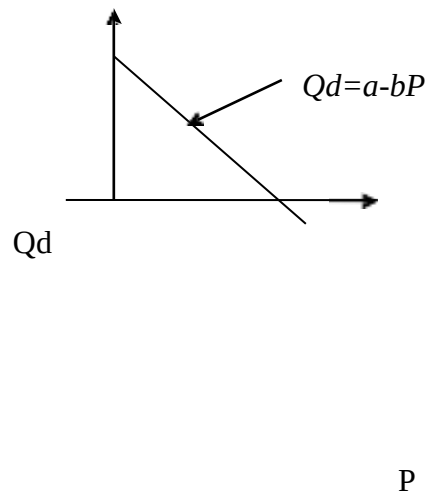
(a) Linear demand and supply functions and equilibrium of the Market

The portions of linear demand and supply functions that occur in the first quadrant are relevant to economic analysis. This is because the quantities we deal with in economics are non negative.

Demand function

A simple demand function such as $Q_d = a - bP$: $a, b \geq 0$, represents the relationship

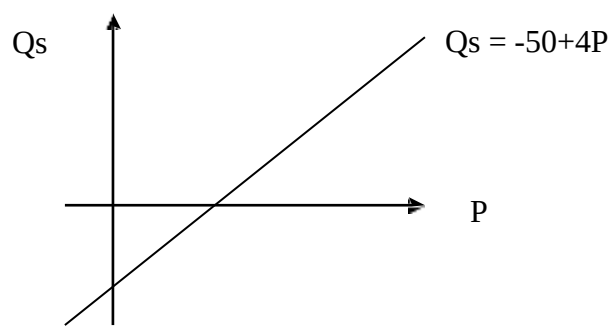
between Q_d , the quantity demanded of a commodity and P , the corresponding price of the commodity. Because the slope coefficient of the function is negative ($-b$), the demand curve will slope downwards to the right.



The intercept on Q_d axis is positive (a) and that on the P axis is also positive ($\frac{a}{b}$). The following characteristics can therefore be deduced about the demand function; “as prices of the commodity in question rise, the quantity of that commodity demanded will fall, and conversely, as price fall, quantity demanded will rise”.

Supply function

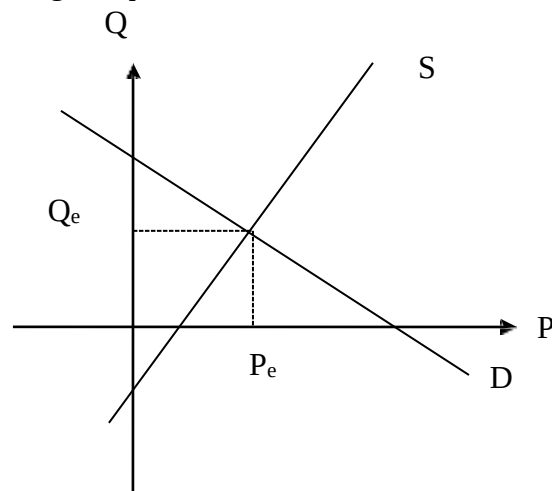
Unlike the demand function, the supply function slopes upwards to the right. In other words it has a positive slope coefficient. For example, $Q_s = -50 + 4P$. The function gives the quantity Q_s that will be supplied at various price levels, P .



We can deduce the characteristics of the supply from the supply function. “The quantity of commodity supplied will increase with increase in the price of the commodity. Conversely, as price falls, so will the quantity supplied also fall”.

Equilibrium of market model

The interaction between supply and demand in the market will generally give rise to equilibrium. Equilibrium in this case, refers to the condition when the quantity supplied and the quantity demanded are equal. This is given by the point of intersection of the demand and supply functions. Equilibrium price and quantity are therefore, the levels of Price and quantity pertaining to equilibrium.



Computation of equilibrium

Given $Q_d = 30 - P$

$$D_s = -20 + 3P \text{ at}$$

equilibrium, $Q_d = Q_s$; hence

$$30 - P = -20 + 3P$$

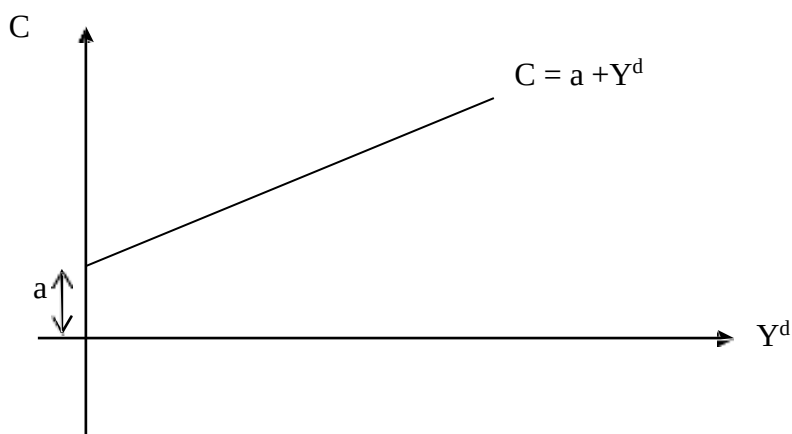
$$50 = 4P; P_e = 12.5 \text{ and } Q_e = 30 - 12.5 = 17.5$$

(b) Consumption function

$$C = a + bY^d; a \geq 0, 0 \leq b \leq 1$$

The expression tells us that consumption expenditure, C , has two main components. The first component represented by a , does not depend on disposable income, Y^d . It is referred to as autonomous consumption. The second part represented by bY^d depends on disposable income.

Since the function has a positive slope, the graph slopes upwards to the right.



The positive slope means that as disposable income increases, so too will expenditure on consumption increase. The slope coefficient b , is known as the marginal propensity to consume (MPC) out of disposable income. It measures the proportion of additional income that is allocated to consumption.

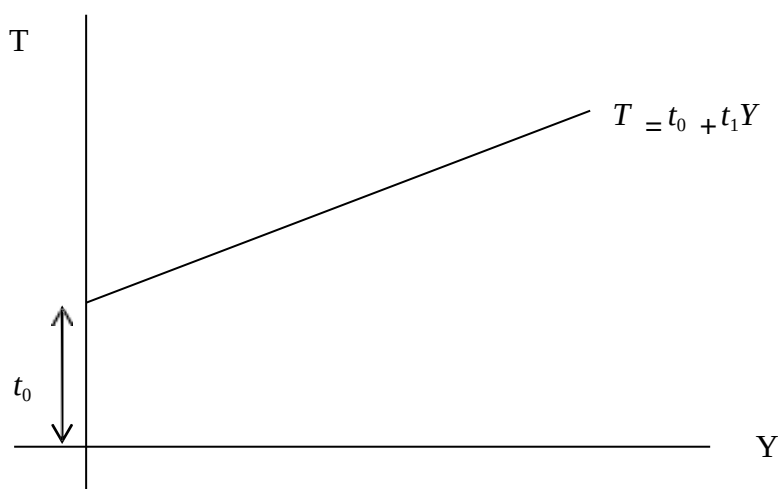
(c) Tax function

The tax function defines the relationship between total tax T , and income Y . in the linear tax function it is assumed that the total tax varies linearly with variations in income, Y , and can be represented as follows $T = t_0 + t_1 Y$: $t_0 \geq 0$ and $0 \leq t_1 \leq 1$; where t_0 and t_1 are constants.

the function implies that total tax T , is made up of two parts: the non income tax represented by t_0 , which does not depend on income and the income tax represented by $t_1 Y$. The latter depends on income.

The slope coefficient t_1 is known as the tax rate. it is positive in sign, hence the graph

slopes upwards to the right. The tax rate measures the amount of any change in income that is diverted into tax payments. At zero income the amount of tax collected is given by t_0 .



Exercise



Given the following consumption and tax functions

$$C = \alpha + \beta Y^d \quad T = tY \text{ where } Y^d = Y - T$$

- (i) Express C as a function of Y
- (ii) Find the level of C when $Y = 500$
- (iii) Find the level of C when $Y = 500$; $\alpha = 50$; $\beta = 0.8$ and $t = \frac{1}{4}$

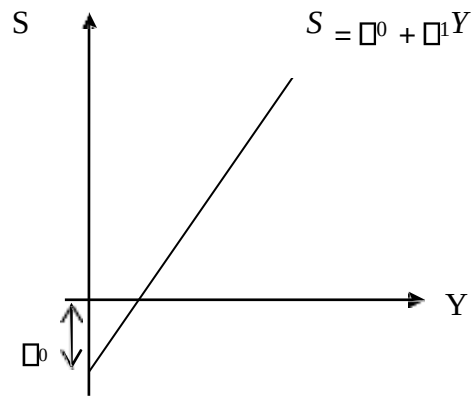
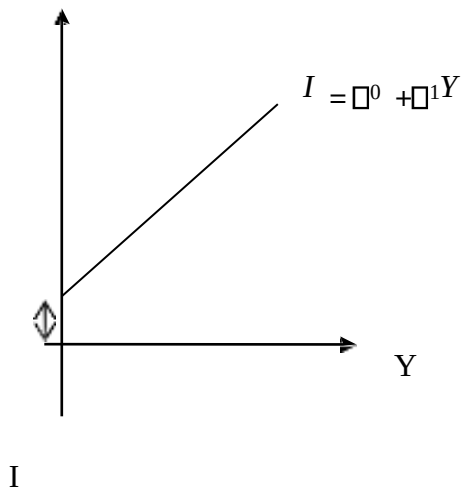
(d) Investment and savings

In economic theory, both investment and savings are assumed to be functions of income and rate of interest. In this section we will assume that investment and savings are functions of income only and are represented by the following equations $I = \alpha_0 + \alpha_1 Y$

$$S = -\alpha_1 + \alpha_1 Y \text{ where } \alpha_0 \geq 0, \alpha_1 \geq 0; 0 \leq \alpha_1 \leq 1; 0 \leq \alpha_1 \leq 1$$

Each of these functions has two parts; the part that is independent of income and the part that is dependent on income. In the investment function, α_0 does not depend on income and is referred to as autonomous investment, and $\alpha_1 Y$ is induced by income. In the savings function, α_0 is independent of income while $\alpha_1 Y$ varies with variations in income.

The slope coefficients of both functions are positive, such that with increase in income both savings and investment increase. The slope coefficient of the investment function is referred to as the Marginal Propensity to invest (MPI). It measures the proportion of any additional income that is diverted to investment. In the case of savings the slope coefficient measures the proportion of any additional income that is diverted into savings and is referred to as Marginal propensity to save.



I_0

(e) Import and export functions

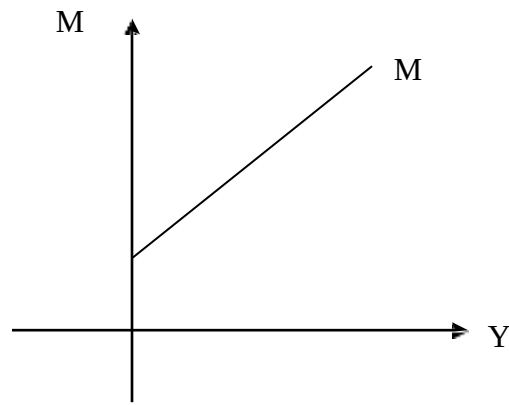
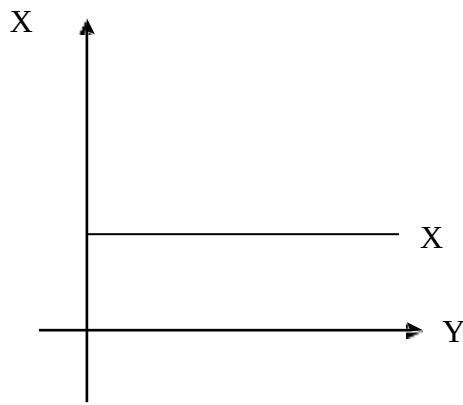
Imports (M) and exports (X) are two components of the national income model that describes the trade that exist between a country and the rest of the world. Both can be represented as functions. Where as imports are assumed to be an increasing function of income, Y, exports are considered to be wholly autonomous of income, thereby giving rise to a constant function.

$$M = m_0 + m_1Y$$

$$X = \bar{X}$$

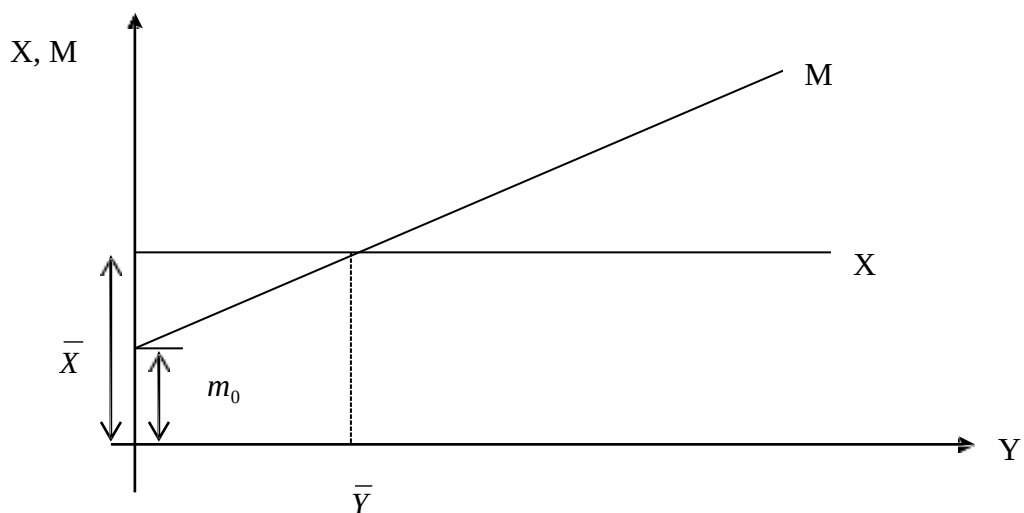
The import function consists of autonomous imports and induced imports. Autonomous imports (m_0) do not depend on income while induced imports (m_1Y) depend on income. The coefficient m_1 is referred to as the marginal propensity to import and it measures the proportion of any additional income that is allocated to imports.

The export function implies that exports (X) is constant and is equal to $\bar{X}$



Equilibrium balance of trade

Balance of trade is said to be in equilibrium when exports are equal to imports.



In the diagram above, balance of trade occurs at the level of income $\bar{Y}$. This is the level of income at which $X=M$.

Below $\bar{Y}$, the level of exports is higher than that for imports. When this happens in a country we say that the country has a surplus balance of trade. When $Y > \bar{Y}$, imports exceed exports and the balance of trade is said to be in deficit.

Exercise

An economy is defined by;

$$Y = C + I + G + X - M$$

$$C = c_0 + c_1 Y$$

$$I = i_0 + i_1 Y$$

$$M = m_0 + m_1 Y$$

$$G = G_0$$

$$X = X_0$$

- (i) Name the endogenous and exogenous variables in this model
- (ii) What is the difference between c_0 and m_0 on the one hand and c_1 and m_1 on the other hand.
- (iii) What is the balance of trade position in the economy when $X > M$; $X < M$ and when $X = M$
- (iv) Find the value of C in terms of $c_0, c_1, i_0, i_1, m_0, m_1, X_0$ and G_0 . denote the value of C computed by $\bar{C}$.

CHAPTER 5

**5.0 NON-LINEAR FUNCTIONS AND MULTIVARIATE FUNCTIONS****Types of functions**

Functions can be categorized into two main classes namely; algebraic and non-algebraic functions. In order to draw the distinction between algebraic and non-algebraic functions, we need to know the meaning of polynomial functions.

The word polynomial means “multiterm”, therefore a polynomial is a multiterm function which is expressed in general form as $y = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$

Several sub categories of polynomials can be generated from the general form by varying the value of n.

- (i) If $n = 1$; we have a, linear function, which refers to polynomial of first degree. e.g.
 $y = a_0 + a_1x$
- (ii) If $n = 2$; we have a quadratic which is a second degree polynomial e.g. $y = a_0 + a_1x + a_2x^2$
- (iii) If $n = 3$; we obtain a cubic function, which is a third degree polynomial. e.g.
 $y = a_0 + a_1x + a_2x^2 + a_3x^3$

Algebraic functions

An algebraic function is any function expressed in terms of polynomials, roots of polynomials or both. Therefore linear, quadratic and cubic polynomials are all algebraic functions.

(a) The general rational functions

The general rational function is a function defined in terms of a ratio of two polynomials. For example $y = \frac{2x^2 + 3x + 5}{18x + 7}$

(b) The circle

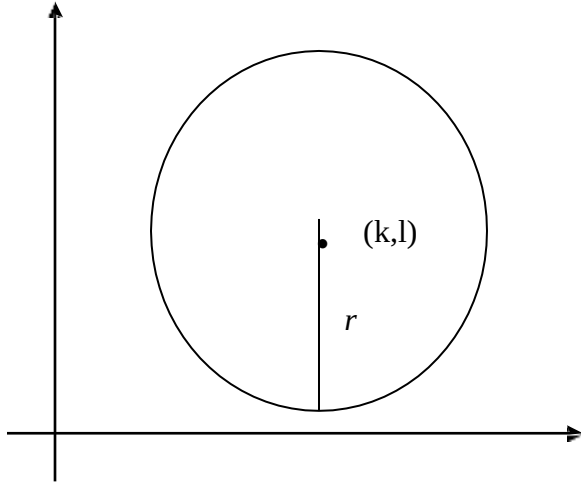
A circle is a locus of points with a fixed distance from the center. The distance from the points from the center is referred to as the radius of a circle. The circle is represented by the following standard form $(x - k)^2 + (y - l)^2 = r^2$; where (k, l) is the centre of the circle and r the radius of the circle.

The standard formula for determining if a circle exists is as follows.

If $r^2 \geq 0$ - a circle exists

If $r^2 = 0$ -no circle exists. The locus forms a point

If $r^2 < 0$ -no real circle exists and the radius is imaginary



Exercise



33

Write the expression $x^2 + y^2 - 4x - 4y = 5$ in the standard form for a circle and check whether or not the loci formed by the expression give rise to a circle

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(c) The parabola

The parabola can be written in one of the following forms

$$Ax^2 + Bx + Cy + D = 0 \text{ --- (1)}$$

or

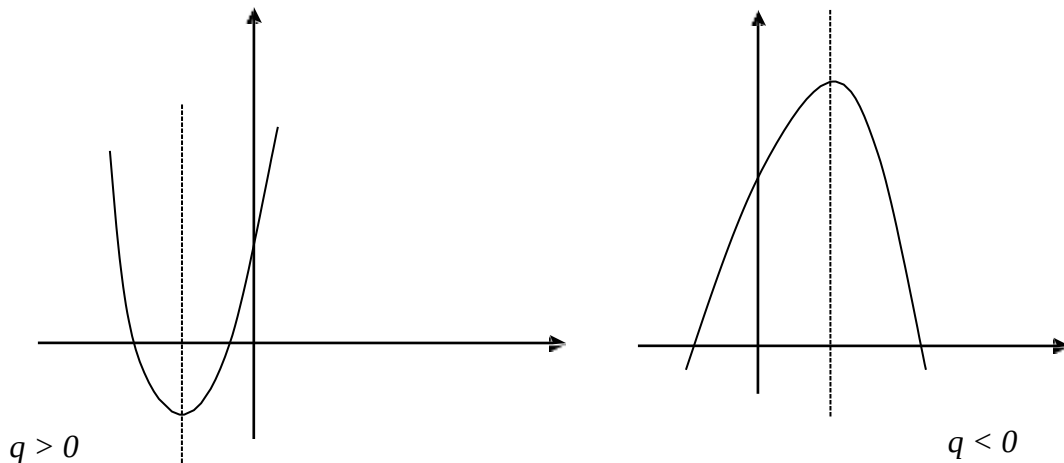
$$Ay^2 + Bx + Cy + D = 0 \text{ --- (2)}$$

The form in (1) represents a parabola whose axis is parallel to the y-axis and the form in (2) represents a parabola whose axis is parallel to the x-axis.

The corresponding standard form for the first parabola is given by

$$(x - k)^2 = 4q(y - l) \text{ Where } (k, l) \text{ is the vertex of the parabola.}$$

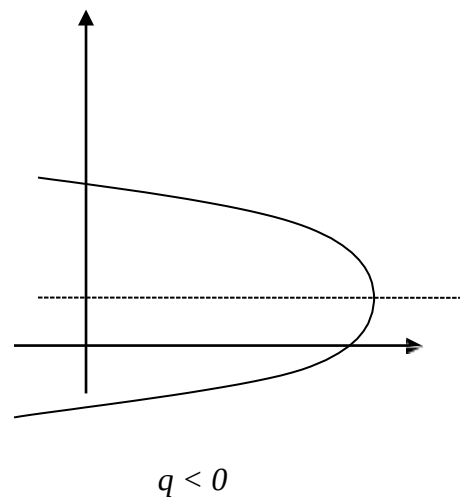
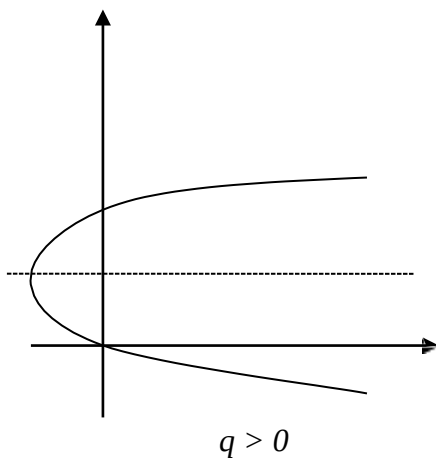
In graphical form the parabola can take either of the following shapes



The standard form for the second parabola is as follows

$$(y - l)^2 = 4q(x - k) \text{ Where } (k, l) \text{ is the vertex. The graphs are as follows}$$

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Exercise



Construct the appropriate standard form for parabola corresponding to the expression $X^2 - 10x + 2y + 31 = 0$ and sketch the corresponding parabola.

(d) The hyperbola

There are several types of hyperbolic functions. However, our main interest is on those with perpendicular asymptotes, referred to as rectangular or equilateral hyperbola. the hyperbola has the standard form $xy = k$, or $y = \frac{k}{x}$; where k is a constant.

The hyperbola has the following characteristics;

- Its asymptotes are the coordinate axes
- It is symmetric about the origin which is also its centre
- If $k > 0$; the hyperbola will have its branches in the first and the third quadrants, while if $k < 0$; it will have its branches in the second and fourth quadrants - y and x are inversely proportional to each other.

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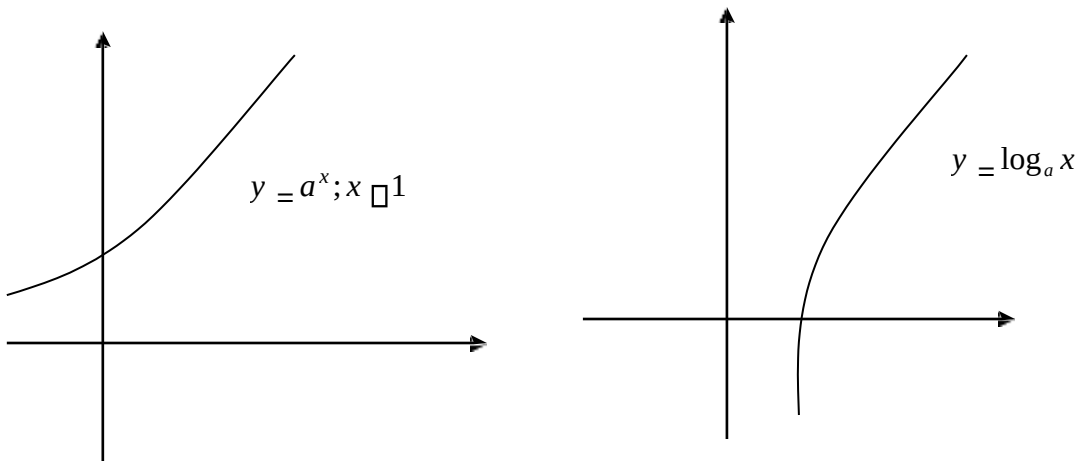
u.

$$k > 0$$

$$k < 0$$

The non algebraic functions

These include exponential functions such as $y = a^x$ in which the independent variable x , appears as an exponent, and logarithmic functions such as $y = \log_a x$. These give the following types of graphs



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Economic applications of non linear functions



Revenue Functions

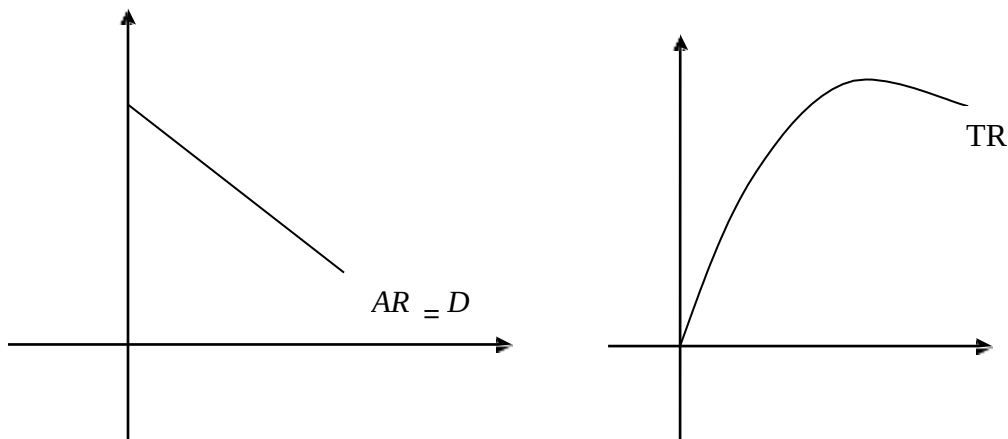
Total revenue (TR) is defined as quantity of output times the price per unit of output thus
 $TR = Q.P$

From the general demand function, we know that $P = P(Q)$. this implies that $TR = Q.P(Q)$

The corresponding Average revenue (AR) function is given by

$$AR = \frac{TR}{Q} = \frac{Q.P(Q)}{Q} = P(Q)$$

This analysis implies that the Average revenue and demand curves for a product are identical.



Numerical Computation

If the Average revenue is given by the linear function $Q + 3p = 6$; the total revenue function can be obtained as follows.

- First express the average revenue function in terms of P as a function of Q $P(Q)$

$$= \frac{6 - Q}{3}$$

Therefore $\frac{d}{dQ} (6 - Q) = -1$

$$TR = Q \cdot P(Q) = Q(6 - Q) = 6Q - Q^2$$

$$\frac{dTR}{dQ} = 6 - 2Q = 0 \Rightarrow Q = 3$$

5.2 Cost Functions



The total cost incurred in producing a commodity consists of *fixed costs* and *variable costs*.

$$TC = FC + VC$$

As the level of output varies the level of fixed costs remain constant. Fixed costs include salaries of persons necessary even during periods of temporary breakdowns, interests and insurance. Unlike the fixed costs, variable costs vary with the level of output. These include costs of materials, and direct labour.

The total costs are normally expressed as a function of Q only. In the general form it is expressed as

$$TC = F(Q)$$

The function used commonly to present Total Cost function is the cubic function.

$$TC = a_0 + a_1 Q + a_2 Q^2 + a_3 Q^3 \quad (a_0, a_1, a_2, a_3 \geq 0, a_3 > 0)$$

The cubic total cost function represents the sum of fixed costs and Variable costs. $FC = a_0$

$$VC = a_1 Q + a_2 Q^2 + a_3 Q^3$$

From these expressions Average total costs (ATC), the Average Fixed Cost (AFC) and the Average Variable Costs (AVC) are calculated by dividing the TC, FC and AV by Q. thus

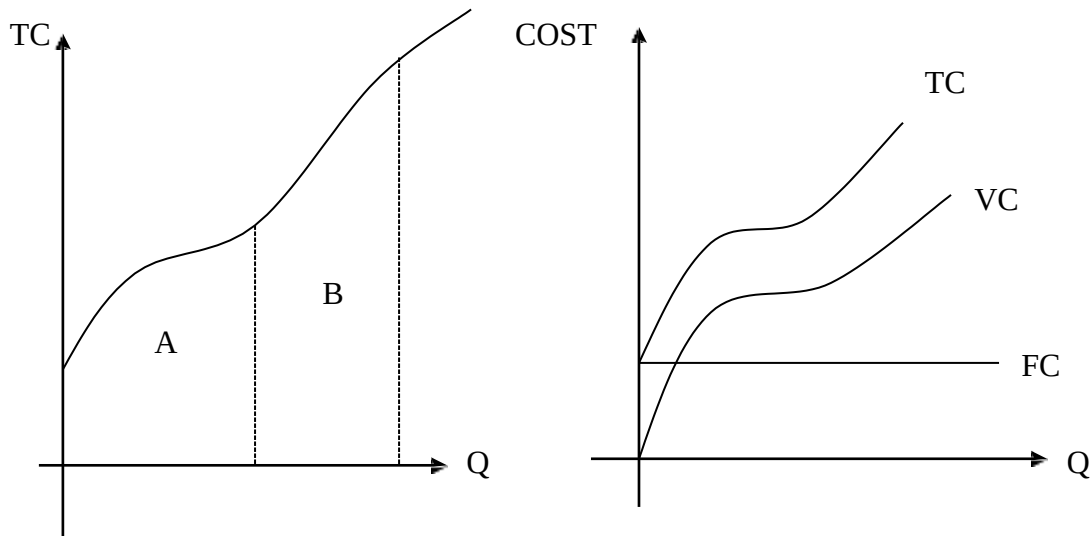
$$ATC = \frac{TC}{Q} = \frac{a_0}{Q} + a_1 + a_2 Q + a_3 Q^2$$

$$AFC = \frac{FC}{Q} = \frac{a_0}{Q}$$

$$AVC = \frac{VC}{Q} = \alpha_1 + \alpha_2 Q + \alpha_3 Q^2$$

Thus the ATC is shown to be the sum of AFC and AVC

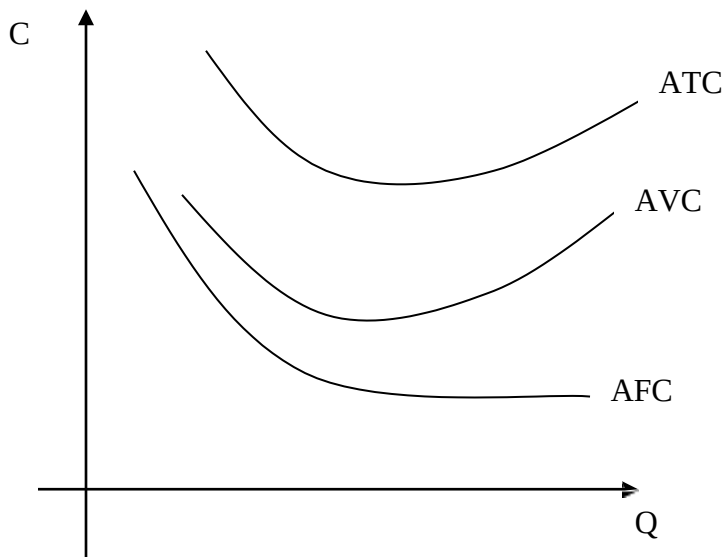
These cost functions can also be represented on graphs as follows;



The following are the properties displayed by the TC function

1. When the output level is zero, TC will be positive, and the relevant cost at this point is the fixed costs.
2. As the level of output increases, Total costs also increase
3. Over some limited range, total cost will increase at a decreasing rate (section A). Over this range the curve is concave downwards
4. Beyond some point (section B), the costs will increase at an increasing rate, thereby giving rise to a curve that is concave upwards

The average costs curves



Exercise



1. Determine the Average Fixed Cost and the Average Variable cost function for a firm whose total cost function is defined by

$$TC = \square + \square_1 Q + \square_2 Q^2 + \square_3 Q^3$$

2. The following are Average total cost and Average revenue functions

$$3 - \frac{15}{Q} - Q = 0$$

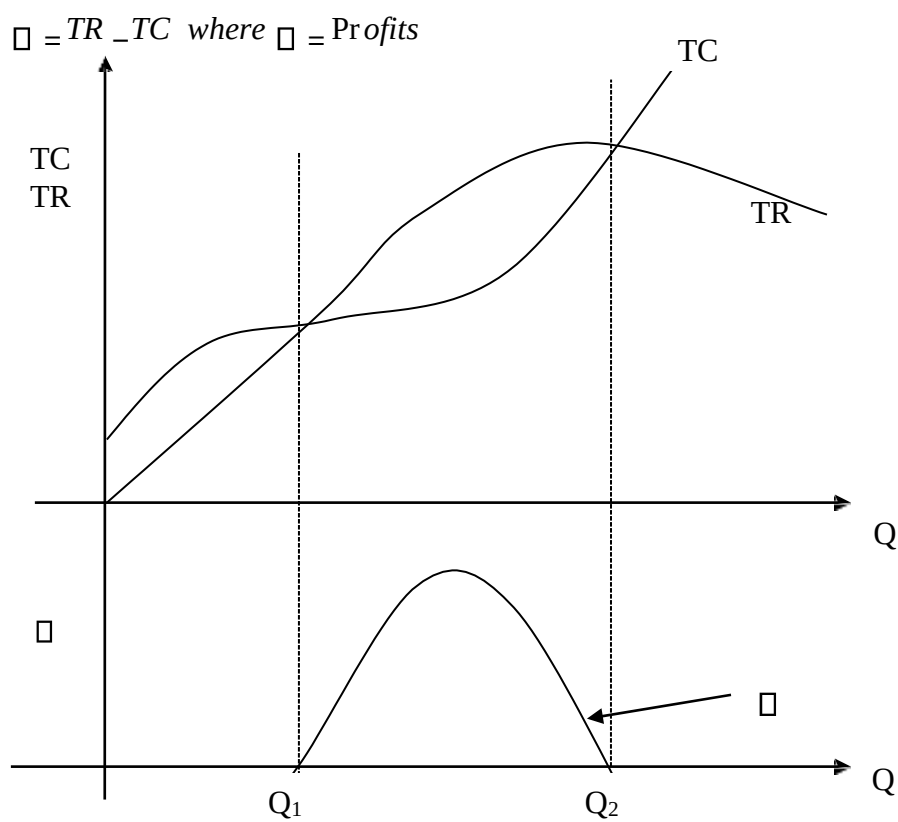
$$AR + \frac{4}{9} Q = 12$$

Determine the Total costs, Average variable Costs, Average fixed costs and total revenue functions.

5.3 Profit Functions



The profit of a firm is defined as the difference between the total revenue and the total costs of the firm,

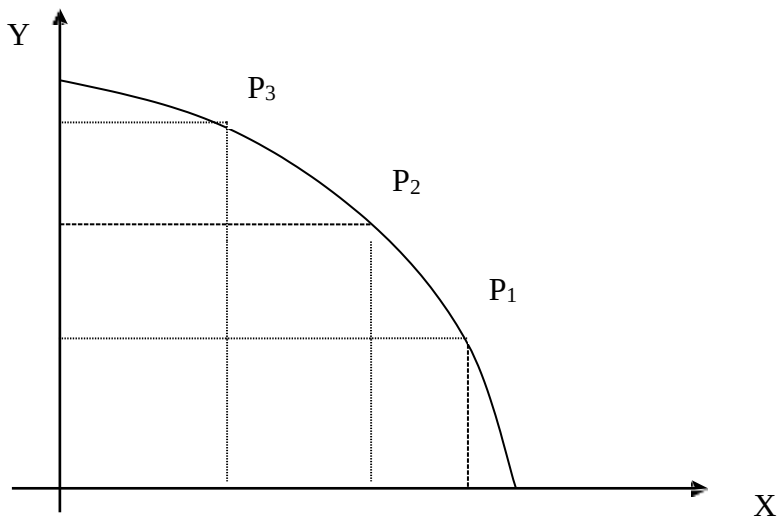


As indicated in the above figure, the level of profit changes with changes in Q . between $Q = 0$ and Q_1 , profit is negative, at Q_1 profits is zero. After Q_1 , profit rises and reaches a maximum between Q_1 and Q_2 , then declines to zero at Q_2 and becomes negative again beyond Q_2 .

5.4 The product transformation curves



Production transformation curves are sometimes referred to as production possibility (PP) curve. It is a locus of combinations of output levels of products X and Y which can be obtained by the use of a given level of input.

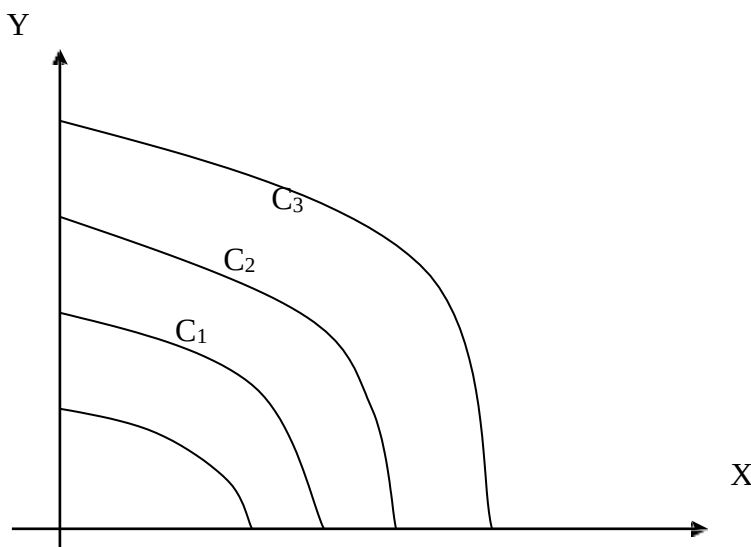


The PP curve has the characteristic that it is concave to the origin, and that as the quantity of one product increases the quantity of the second product will concurrently be decreasing

as we move from point p_1 to p_2 then to p_3 , along the production possibility curve, the quantity of y will be increasing while the quantity of x will be decreasing. Thus we will be transforming the production of x into y . by producing less x in favour of production of more and more of y .

PP curves usually occur in a group. in such a group of family each one of the curves correspond to a different level of input and the further away from the origin a curve is the higher the level of input to which it corresponds.

Thus in the figure below, input level corresponding to C_1 is lower than that corresponding to C_2 or C_3



Functions of more than one independent variable (multivariate functions)



The concept of a function can readily be extended to two or more independent variables, e.g.

$$z = t(x, y)$$

$$= g(x, w, y)$$

In the first function a pair of x , and y values will uniquely determine on value of the dependent variable z . in the second function, x , w and y values will uniquely determine one value of z .

Multivariate functions are difficult to plot on two dimensional surface because they need more axes than we can plot on a two dimensional plane.

Economic applications of Multivariate functions



Multivariate functions are very important in the description of many problems in economic analysis.

(i) Consumer demand functions.

Suppose the quantity q that a consumer demands depends on price P_1 , of the commodity, the income, Y of the consumer and the price P_2 of some related good, then the general way of expressing the information is $q = f(P_1, P_2, Y)$ or in a more particular/ specific form as

$$q = \alpha_0 + \alpha_1 P_1 + \alpha_2 P_2 + \alpha_3 Y$$

From these we may want to know the effect on q of change of a fall in P_1 or a rise in Y . Given the signs of the parameters $\alpha_0, \alpha_1, \alpha_2$, and α_3 of the function, we can easily answer such questions.

(ii) Demand and revenue functions of related commodities

The demand and revenue functions of related commodities are multivariate functions. The demand functions of two related commodities Q_1 and Q_2 with

prices P_1 and P_2 can be represented by

$$Q_1 = Q_1(P_1, P_2)$$

$$Q_2 = Q_2(P_1, P_2)$$

The specific forms of such functions would generally permit the analyst to obtain the corresponding inverse functions

$$P_1 = P_1(Q_1, Q_2)$$

$$P_2 = P_2(Q_1, Q_2)$$

These inverse functions are also the average revenue functions for the commodities Q_1 and Q_2 . Thus

$$AR_1 = P_1 = P_1(Q_1, Q_2)$$

$$AR_2 = P_2 = P_2(Q_1, Q_2)$$

The corresponding Total Revenue will be given by

$$TR_1 = P_1 Q_1 = P_1(Q_1, Q_2) Q_1$$

$$TR_2 = P_2 Q_2 = P_2(Q_1, Q_2) Q_2$$

(iii) Utility function

The utility or level of satisfaction derived by a consumer as a result of the consumption of quantities x and y of two different goods X and Y can be represented by a general utility function

$$U = h(x, y)$$

The function has two independent variables and could assume several specific forms such as

$$U = x^2 + 2x + 2y + 4$$

(iv) The production function

The production function is another example of a multivariate function. The general form $Q = Q(K, L)$; implies that output Q of a firm is determined by the amount of labour L and capital K , employed by the firm.

One of the most popular particular forms of the production function in economic analysis is the Cobb-Douglas production function, which can be represented as

$$Q = AK^\alpha L^\beta$$

Such a function can enable us to answer questions on matters to do with marginal products, returns to factors, returns to scale, and factor substitution. For example, the Average product of labour (APL) is obtained by dividing the production function by the total units of labour, and the Average product of capital (APK) is obtained by dividing the production function by the total units of capital employed in the

production process. if the production function is given by the Cobb-Douglas function, then

$$APL = \frac{Q}{L} = AK L^{\alpha-1}$$

$$APK = \frac{Q}{K} = AK^{\alpha-1}L^{\alpha}$$

(v) Joint cost function

The joint cost function of a firm is a function in which the cost of production is expressed as a function of two or more different commodities produced by the firm, and is represented by

$$C = C(Q_1, Q_2)$$

The corresponding average cost functions for commodities Q_1 and Q_2 are given by

$$\frac{C(Q_1, Q_2)}{Q_1} \text{ and } \frac{C(Q_1, Q_2)}{Q_2} \quad AC_1 =$$

Equations and inequalities

CHAPTER SIX

6.0 LINEAR SIMULTANEOUS EQUATIONS



Introduction



An economic model will generally have two or more interrelated parts or sub models. Each one of the sub models is represented by an equation designed to describe the structure of the sub model. Since the sub models are interrelated the equations representing them will also be interrelated and will normally constitute a system of simultaneous equations.

A system of simultaneous equations can either be made up of linear or non linear equations or a combination of the two

Linear simultaneous equations



Two or more equations will form a system of linear simultaneous equations if such equations be linear in the same two or more variables. For example

$$2x + 6y = 23$$

$$10x + 7.5y - 5z = 10$$

or

$$4x + 7y = 10$$

$$15x + 12.5y + 17.5z = 42.5$$

Solutions of a system of linear simultaneous equations

The solution of a system of simultaneous equations is a set of values of variables which simultaneously satisfy all the equations of the system. In general, the solution to a set of n linear simultaneous equations in n variables is a set of n values, one for each of the n variables, which satisfy all the equations of the system.

There are three different methods of solving linear simultaneous equations;

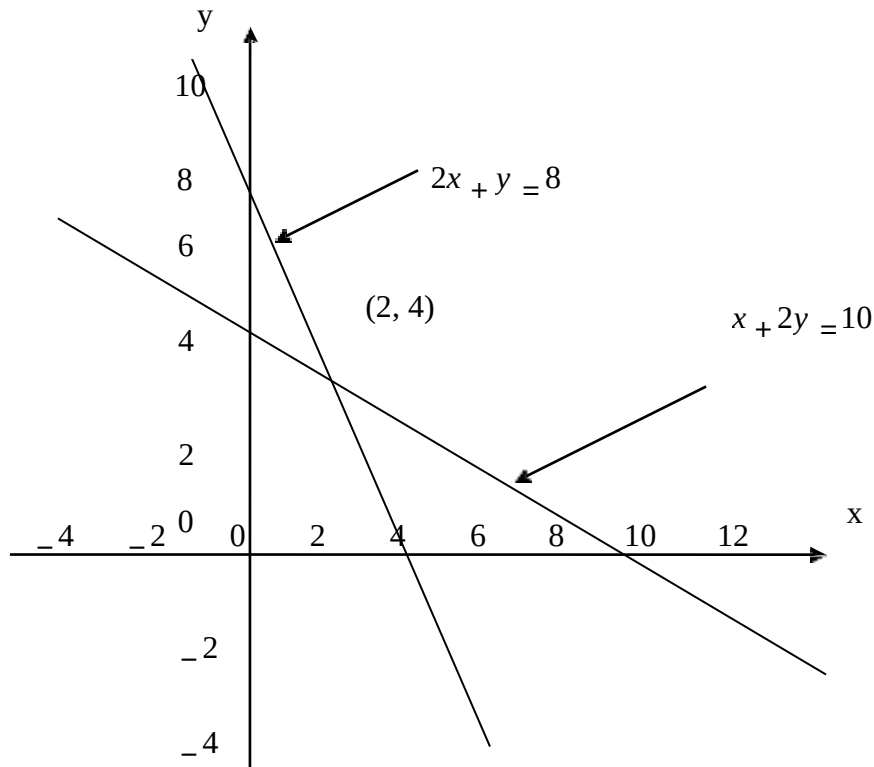
- Solution by elimination
- Solution by substitution
- The graphical method

A system of linear simultaneous equations may have no solution, it may have exactly one solution referred to as a unique solution, or it may have two or more solutions. The case where there is no solution is the case of an *inconsistent* system. The other cases are for *consistent* system.

The graphical method of solving simultaneous equations



The graphical technique of solving a system of linear equations consists of drawing the graphs of the equations of the system on the same rectangular coordinate system. the coordinates of the point of intersection of the equations of the system would then be the solution.



$$2x + y = 8$$

This means that the solution to the system is $(2, 4)$ $x + 2y = 10$

That is $x = 2$ and $y = 4$, defined by the point of intersection of the equations.

The elimination method



solution of a system of simultaneous equations require that each variable be eliminated in turn by making the absolute value of its coefficient equal in the equations of the system, and then adding or subtracting the equations.

Consider a system of two linear equations

$$2x - 3y = 8 \quad \text{---(1)}$$

$$3x + 4y = -5 \quad \text{---(2)}$$

Step 1: Multiply (1) by 3 : $6x - 9y = 24 \quad \text{---(3)}$

Multiply (2) by 2 : $6x + 8y = -10 \quad \text{---(4)}$

Subtract (4) from (3): $-17y = 34 \quad \text{---(5)}$

Solve (5) for y: $y = -2 \quad \text{---(6)}$

Step 2: Multiply (1) by 4: $8x - 12y = 32 \quad \text{---(7)}$

Multiply (2) by 3: $9x + 12y = -15 \quad \text{---(8)}$

---(8) Add (7) to (8): $17x = 17 \quad \text{---(9)}$

---(9)

solve for x from (9): $x = 1 \quad \text{---(10)}$

Solutby substitution



Consider a system of two equations

$$2x - 3y = 8 \quad \text{---(1)}$$

$$3x + 4y = -5 \quad \text{---(2)}$$

Step 1: From equation (1) find an expression for x

$$2x - 3y = 8; \quad 2x = 3y + 8; \quad x = \frac{3}{2}y + 4 \quad \text{---(3)}$$

Step 2: Substitute the value of x in (3) into (2)

$$3x + 4y = -5 : \quad 3\left(\frac{3}{2}y + 4\right) + 4y = -5 : 4.5y + 12 + 4y = -5$$
$$\square \quad \square$$
$$\square 2 \quad \square$$

$$8.5y + 12 = -5; 8.5y = -17, - - - - - (4)$$

Step 3: Solve equation (4) for y

$$8.5y = -17$$

$$y = -2$$

Step 4: substitute the value of $y = -2$ into (1) or (2) $2x$

$$-3y = 8 : 2x - 3(-2) = 8 : 2x + 6 = 8; 2x = 2 \text{ hence}$$

$$x = 1$$

NB: note that the solution is similar to the solution we obtained by elimination.

Exercise



Solve by elimination and substitution techniques, the following system of linear equations $x + y = 3$ $x - 2y = 9$

Economic Application of linear simultaneous equations



Systems of simultaneous equations are useful in **equilibrium analysis**.

In market models that involve one commodity alone, quantity demanded (Q_d) and quantity supplied (Q_s) are functions of the price of that commodity alone. Equilibrium in such markets are referred to as partial market equilibrium, where the word 'partial' refers to the fact the commodity is assumed to depend on its price alone and independent of the price of other commodities

In the real world, most commodities are related to other commodities as substitutes or complements. Hence a more realistic way of presenting the demand and supply functions of a commodity should take into consideration the effect of price of related commodities. The existence of several interdependent commodities give rise a general market as opposed partial market equilibrium. If such general market equilibrium exists, then the quantity demanded of in each market will be equal to the quantity supplied in the markets.

Elements of a market model

Consider a single commodity market model.

Only three variables are included in the model: the quantity demanded (Q_d), quantity supplied (Q_s) and the price of the commodity, P .

Three basis assumptions are adopted in the construction of the model

- Q_d is a decreasing function of P
- Q_s is an increasing function of P
- For equilibrium to hold Q_s and Q_d must be equal

These assumption when translated into mathematical form will give two behavioral functions and equilibrium condition, given as follows

$$Q_d = a + bP \quad a, b \leq 0$$

$$Q_s = m + nP \quad m, n \geq 0$$

$$Q_s = Q_d$$

The principal aim in equilibrium analysis is to find a set of values of the endogenous variables that will satisfy the equilibrium condition of the model.

In the model above there are three endogenous variables: Q_s , Q_d and P . the equilibrium values of these variables are denoted $\bar{Q}_d$, $\bar{Q}_s$ and $\bar{P}$. At equilibrium $\bar{Q}_d = \bar{Q}_s$ and by

substituting $\bar{P}$ into the behavioral equations, the two equations become equal.

Exercise



A single commodity market model is given by

$$Q_d = 18 - 2/3P$$

$$Q_s = -6 + 3/4P$$

Where Q_d and Q_s are quantities demanded and supplied and P is the price. Calculate the equilibrium price and quantity for the market.

The two commodity market model

When a market has more than one interrelated commodities, equilibrium in such a market require that quantity demanded and quantity supplied be equal for each one of the commodities involved. That is

$$Q_{di} = Q_{si} : i = 1, 2, \dots, n$$

Consider the following two commodity market model in which both demand and supply functions are assumed to be linear

$$Q_{d1} = 8 - 2P_1 + P_2$$

$$Q_{s1} = -5 + 3P_1$$

$$Q_{d1} = Q_{s1}$$

$$Q_{d2} = 16 + P_1 - P_2$$

$$Q_{s2} = -1 + 2P_2$$

$$Q_{d2} = Q_{s2}$$

The equilibrium solution of the model is determined by making use of the two equilibrium conditions, which generates a system of simultaneous equations

$$\begin{aligned} 8 - 2P_1 + 2P_2 &= -5 + 3P_1 & 5P_1 - P_2 &= 13 \\ 16 + P_1 - P_2 &= -1 + 2P_2 & P_1 - 3P_2 &= -17 \end{aligned}$$

Solving the system yields $P_1 = 4$; $P_2 = 7$

The equilibrium quantities are obtained by substituting $P_1 = 4$; $P_2 = 7$ into the demand or supply equations

$$Q_{d1} = Q_{s1} = Q_1 = 7$$

$$Q_{d2} = Q_{s2} = Q_2 = 13$$

The national income model

The simple Keynesian national income model can be represented by:

$$Y = C + I_0 + G_0$$

$$C = \alpha + \beta Y$$

The variables Y and C are endogenous, I_0 and G_0 are exogenously determined

The equilibrium income Y and Consumption can be determined in terms of the parameters α and β and the exogenous variables by solving the system for Y and C

By substituting the equation for C into the equation for Y

$$Y = C + \alpha Y + I_0 + G_0$$

$$(1 - \alpha)Y = C + I_0 + G_0$$

$$Y = \frac{C + I_0 + G_0}{1 - \alpha}$$

Substituting the value of Y in the C equation gives

$$C = C_0 + \alpha \frac{C + I_0 + G_0}{1 - \alpha}$$

The IS-LM model

Equilibrium in the commodity market $I=S$

The total income of an economy that does not engage in foreign trade is expressed as

$$Y = C + S + T$$

The total expenditure on the other hand is expressed as

$$E = C + I + G$$

The equilibrium level of income is that level of income at which the total income for an economy equals the total expenditure for the economy. Since income reflects the cost of goods produced while expenditure reflects the proceeds from sales of goods, then the equilibrium level of income will also be the level of income at which all goods produced are purchased. Hence the condition of equilibrium is derived by equating the two expressions

$$C + S + T = C + I + G$$

Assuming government expenditure equals the tax it collects ($G = T$), we get $S = I$. the condition $S = I$ constitutes the condition for equilibrium in the commodity market.

The IS curve and equilibrium in the goods market

According to Keynesian economics, savings is assumed to be function of income Y, and investment is assumed to be a function of interest rate r. thus $S = s.Y$; $0 \leq s \leq 1$

$$I = A - \alpha r; \alpha \geq 0$$

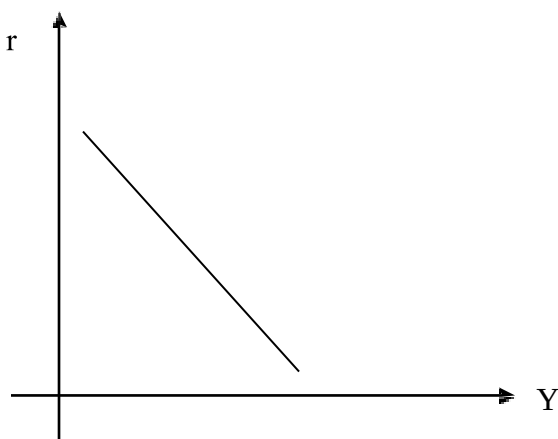
The IS curve is a curve along which savings and investment are equal. Therefore we can easily obtain the function corresponding to the curve by equating the two functions

$$sY = A - \alpha r$$

$$-sY = \alpha r - A$$

$$= -\frac{\alpha}{s} r + \frac{A}{s}$$

$$Y = -\frac{\alpha}{s} r + \frac{A}{s}$$



Thus the IS curve is a linear function of the variables Y and r. The LM curve and equilibrium in the money market

Equilibrium in the money market occurs when the supply of money M_s is equal to Money demand M_d .

According to Keynes money demand has two components; transactions demand for money and Precautionary demand component and the speculative demand component. the assumption is that transactions demand for money is an increasing function of income Y, and speculative demand for money is an inverse function of the rate of interest r. thus $M_d^T = aY$; $a > 0$

$$M_d^S = B - br; B, b > 0$$

At equilibrium $M_s = M_d$, hence $M_s = M_d = M_d^T + M_d^S$

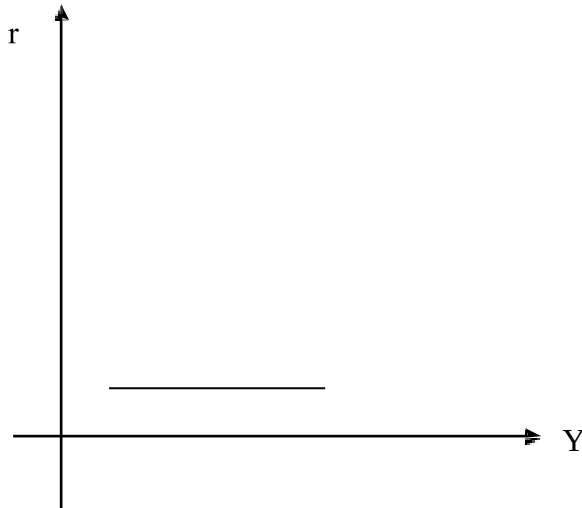
Assume money supply is d, then

$$d = aY + B + br$$

$$br = d - B - aY$$

$$r = \frac{d - B}{b} + \frac{a}{b} Y$$

This equation constitutes the LM curve, the curve along which the demand for money is equal to the supply of money. The function is linear in the two variables Y and r.



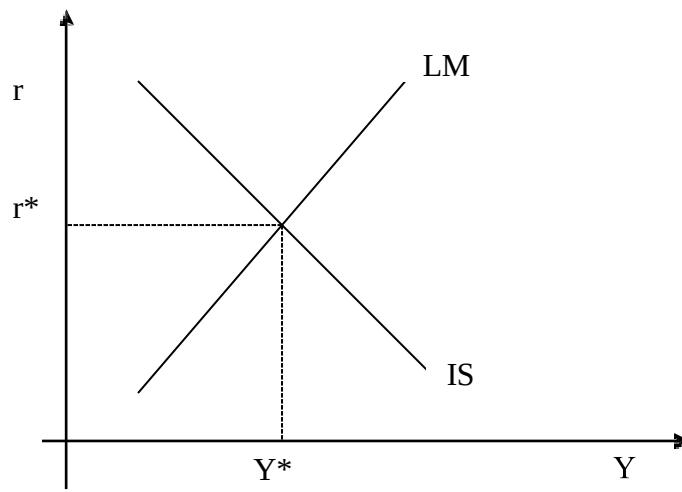
Equilibrium in both goods and Money markets

The simultaneous equilibrium in the goods as well as in the money market will occur when both the equations of the IS and LM hold. That is

$$sY + \square r = A$$

$$aY - br = d - B$$

this constitutes a set of two linear simultaneous equations in Y and r. the equilibrium Y and r are obtained by simultaneously solving the system for Y and r. the equilibrium will therefore occur at the intersection of the IS and the LM curves



Exercise:



You are given the following information about the commodity and money market of a closed economy without government

The commodity market

$$\text{Consumption function: } C = 50 + \frac{2}{5}Y$$

$$\text{Investment function: } I = 790 - 21r$$

The money market

Precautionary and transactions demand for money; $Md^T = \frac{1}{6}Y$

Speculative demand for money: $Md^S = 1200 - 18r$
Money supply: $Ms = 1250$

Find the equilibrium Y and r in the two markets.



7.0 DERIVATIVES AND DIFFERENTIATION



7.1. Rate of change, Limit and Continuity



The concepts of rates of change, limit and continuity are very fundamental in understanding of the principles of differentiation of a process or finding the derivative of a function.

Rate of change/ slope of a function



Rate of change of a function is defined as the change in the value of the dependent variable divided by the corresponding change in the value of the independent variables. The symbol Δ is generally used to represent the notion of change. If a x variable x changes value from x_0 to a new value x_1 the change is measured by the difference

$(x_1 - x_0)$ and such a change is usually represented as Δx

Now consider a linear function

$$y = a + bx$$

When the independent variable x changes from a value of x to a new value $(x + \Delta x)$, y also change from y to a new value $(y + \Delta y)$. Hence the changes will give

$$y + \Delta y = a + b(x + \Delta x)$$

Therefore the increase in the dependent variable y due to the corresponding increase in x (Δx) is given by the difference

$$(y + \Delta y) - y = [a + b(x + \Delta x)] - (a + bx)$$

Algebraically this simplifies to

$$\Delta y = b\Delta x$$

The rate of change of the linear function will therefore be

$$\frac{\Delta y}{\Delta x} = \frac{b\Delta x}{\Delta x} = b$$

The same argument can be adopted to determine the rate of change for non linear functions. For example; suppose $y = x^2$

$$y + \Delta y = (x + \Delta x)^2 = x^2 + 2x\Delta x + [\Delta x]^2 ; \text{ but } x^2 = y$$

$$y + \Delta y = y + 2x\Delta x + [\Delta x]^2$$

$$\Delta y = 2x\Delta x + [\Delta x]^2$$

Dividing both sides by Δx yields

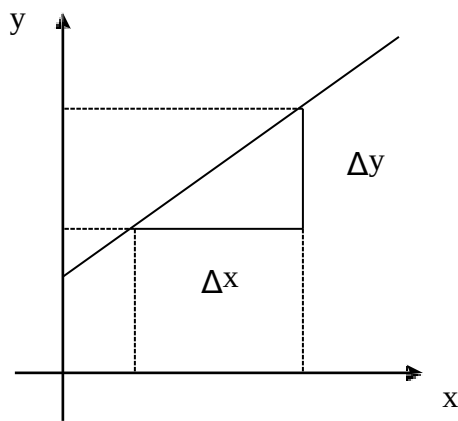
$$\frac{\Delta y}{\Delta x} = 2x + \Delta x$$

The rate of change measures the slope of function. Since to determine the slope of a function within a range we compute change in the vertical distance divided by the change in the horizontal distance; that is

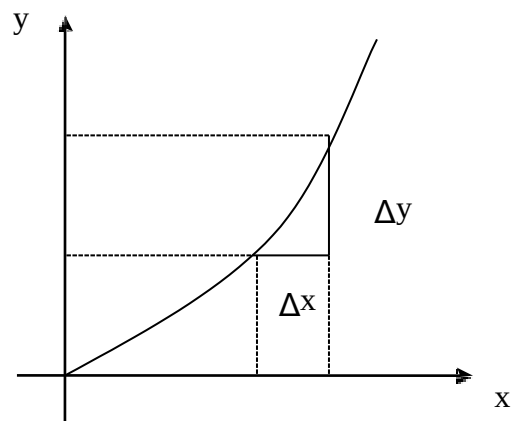
$$\text{slope} = \frac{\text{change in } y}{\text{change in } x} = \frac{\Delta y}{\Delta x}$$

$$\text{in } x \Delta x$$

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a) Linear function



b) Non linear function

According to these results it can be seen that the slope of a linear function is a constant. This means that the slope is the same at every point along the line/ function. However, the slope of a nonlinear function is not constant all over the curve but changes in value from one point to another. This is demonstrated by the expression of the slope which is given as a function of the independent variable. The quantity

$\frac{\Delta y}{\Delta x}$ therefore gives the average rate of change and not instantaneous.

Δx

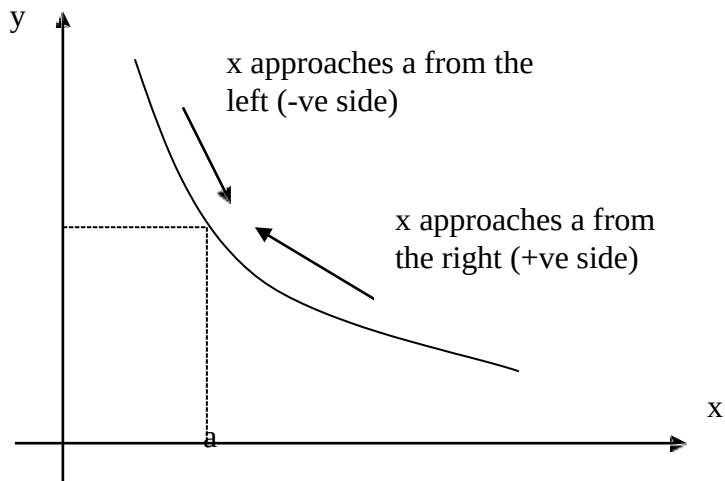
The concept of limits



The concept of the limit of a function tries to answer the question; what value does the dependent variable approach as the independent variable approaches a certain specific value?

Suppose $y = f(x)$ then $\lim_{x \rightarrow a} y = f(a)$ or $\lim_{x \rightarrow a} f(x) = f(a)$

This is read as 'the limit of y as x approaches/ tends to a



The arrows in the diagram show that x approaches a both from the negative (i.e. from

a value smaller than a) and positive side (from a value larger than a).

A limit of a function at a point exists only if the left side (negative) and the right side (positive) limits are equal

Example

Evaluate $\lim_{x \rightarrow 7} (x + 7)$

Solution

$$\lim_{x \rightarrow 7} (x + 7) = (7 + 7) = 14$$

What this implies is that as x comes closer and closer to 7, y comes closer and closer to 14.

Exercise

Evaluate the following

(i) $\lim_{x \rightarrow 0} \frac{x^2}{x} = \frac{0}{0}$ (ii) $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = \frac{0}{0}$ (iii) $\lim_{x \rightarrow 5} 5 + 3x$



A function is continuous in the interval $a = a_1$ and $a = a_2$ if the function has no breaks

in the interval. It is often very necessary to analyze continuity of a function at a point say point $x = a_0$

If the function should be continuous at that point, it must satisfy the following requirements at point $x = a_0$

a) The point $x = a_0$ must be in the domain of the function i.e. " a_0 " must be one of

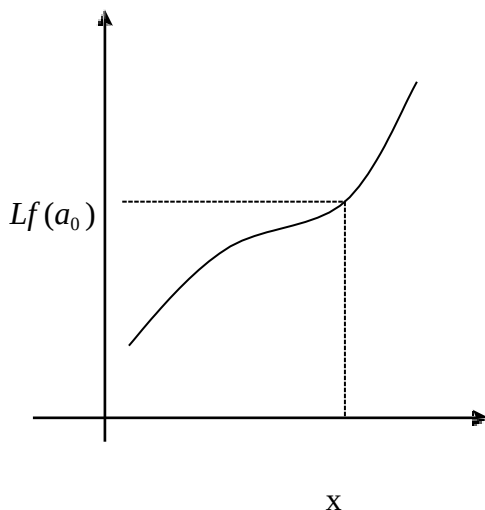
the values that x is permitted to assume.

b) $\lim_{x \rightarrow a_0} f(x)$ must exist. That is the function must have a limit as x approaches

" a_0 "

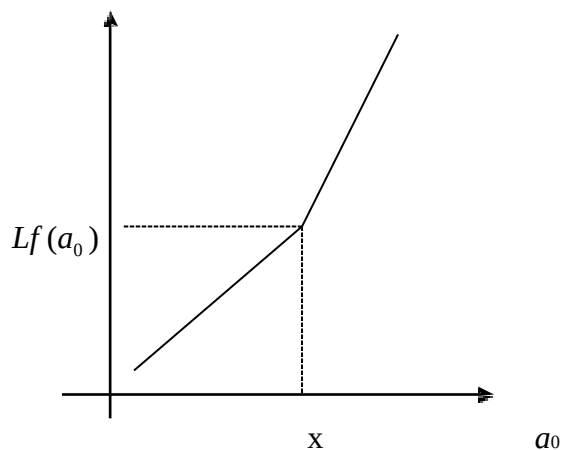
c) $\lim_{x \rightarrow a_0} f(x) = f(a)$ that is as $x \rightarrow a_0$ from the left or from the right, the function should

approach the limit $f(a_0)$



a_0

x



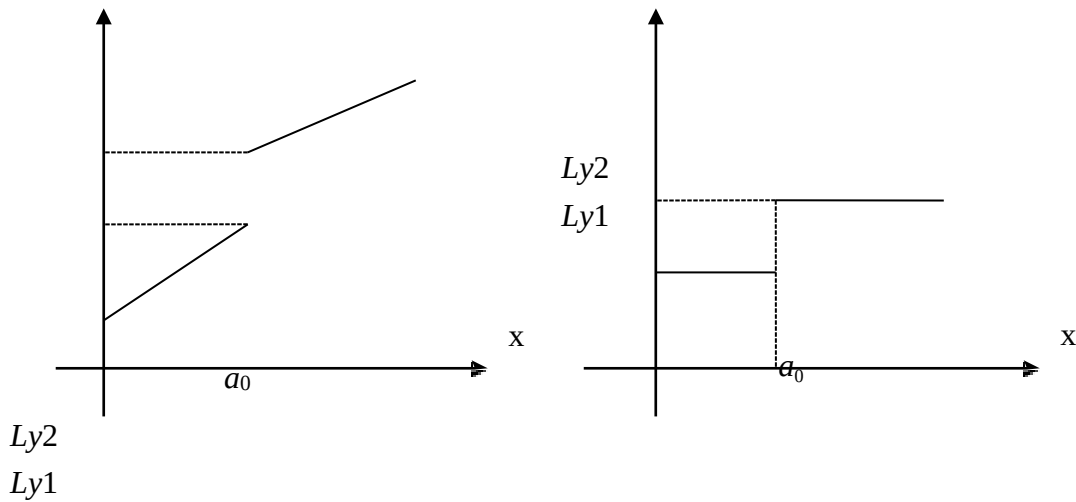
x

a_0

The two functions above are continuous at point a_0 because they satisfy the three requirements for continuous functions

However the functions below are discontinuous at a_0 . The functions have breaks at the points, and therefore do not satisfy the conditions for continuous functions. In these functions although the domain for x includes $x = a_0$, $\lim_{x \rightarrow a_0} f(x)$ do not exist.

The right side limits and the left side limits are different at the point $x = a_0$



Example

Analyze the continuity of $f(x)$ at $x = 12$ given that $f(x) = x + 7$ $0 \leq x \leq 10$

Solution

Since the function is defined in the interval $0 \leq x \leq 10$ only, it is not defined for $x = 12$. We cannot therefore analyze its continuity at $x = 12$ because its value is not in the domain of the function.

Exercise



$$\frac{x^2 - 9}{x - 3}$$

Analyze the continuity of $\frac{x^2 - 9}{x - 3}$ at $x = 3$

$$x - 3$$

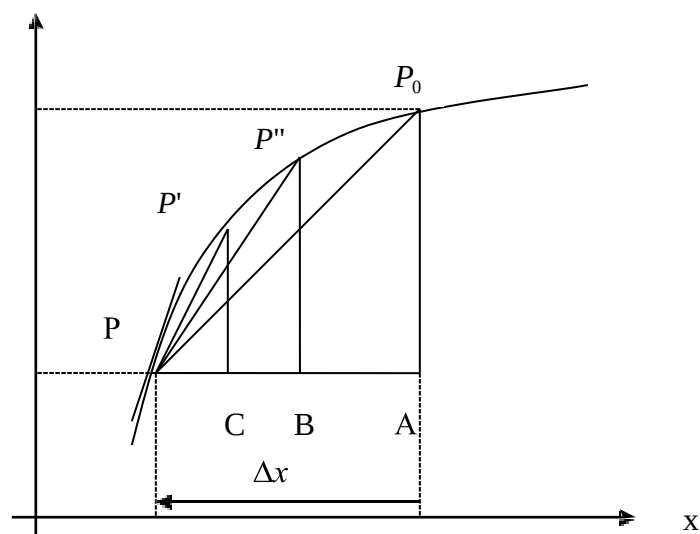
7.2 The derivative and Differentiation



Recall that the slope of a non linear function varies from point to point along the curve.

Since the slope of the curve changes from point to point, we can either calculate an average slope for any arc on the curve, or else calculate the instantaneous slope (point slope) of the curve at any point we choose.

y



Calculating average slope of the chords

For the chord PP_0

Movement along the curve changes the value of x by Δx and the value of y by Δy . The average slope of the arc is

$$\frac{\Delta y}{\Delta x}$$

$$\Delta x$$

For the chord PP''

If between P and P', x changes to $x + \Delta x_1$ and y changes value from y to $y + \Delta y_1$ then the average slope of the arc is given by

$$\frac{\Delta y_1}{\Delta x_1}$$

For the chord PP'

If between P and P' x changes to $x + \Delta x_2$ and y changes value from y to $y + \Delta y_2$ then the average slope of the arc is given by

$$\frac{\Delta y_2}{\Delta x_2}$$

$$\Delta x_2$$

Note that none of these slopes gives the slope of the curve at a particular point. But it can be noticed that as we move along the curve from P_0 towards P, gets smaller and smaller and the slopes of the various chords considered change continuously.

To obtain the slope at point P, it is necessary to reduce to 0. this means that point P_0 that had been moving towards P will have coincided with P and what was originally the chord PP_0 will have turned into a tangent at the point P. the slope of the curve at point P will therefore be

$$\lim_{P_0 \rightarrow P} \frac{\Delta y}{\Delta x} \quad \text{as } \Delta x \rightarrow 0 \text{ and}$$

Thus the slope of the curve at point P is given by

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \text{slope of the tangent at P}$$

Summary



Computation of the slope of a nonlinear function involves the following step

- Determine the average slope of the arc ACB by computing the slope of the chord AB.
- Determine the limiting value of this slope as $\Delta x \rightarrow 0$

Example

For the function $y = x^2$

$$y + \Delta y = (x + \Delta x)^2 = x^2 + 2x\Delta x + [\Delta x]^2; \text{ but } x^2 =$$

$$y \quad y + \Delta y = y + 2x\Delta x + [\Delta x]^2$$

$$\Delta y = 2x\Delta x + [\Delta x]^2$$

Dividing both sides by Δx yields

$$\frac{\Delta y}{\Delta x}$$

$$= 2x + \Delta x$$

$$\Delta x$$

Therefore the slope of the function is

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} 2x + \Delta x = 2x$$

Exercise



Determine the slope of $y = a + 2bx^2$

The derivative of a function $y = f(x)$, denoted by $f'(x)$, is another function defined

$$\text{as } f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} \quad \text{OR} \quad f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

The process of obtaining the derivative of a function is referred to as derivation or differentiation.

A function is said to be differentiable if its derivative exists. Apart from the $\frac{dy}{dx}$ notation $f'(x)$ we also use the notation $\frac{d}{dx}$ to denote the derivative of a function

$y = f(x)$. It is read as the derivative of y with respect to x . Hence

$$\frac{dy}{dx} = \frac{d}{dx} f(x) = f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

The derivative gives the instantaneous rate of change of a function, which is the slope of the function at any specified point on a curve.

Exercise



Find the derivatives of the following functions

(i) $y = 5 + 10x$

(ii) $f(x) = 4 - 3x$

(iii) $y = 13 + 2x^2$

7.4. Rules of Differentiation



Rules of differentiation provides a short cut to differentiation in that one does not always have to compute the limit of the quotient $\frac{\Delta y}{\Delta x}$ as $\Delta x \rightarrow 0$ whenever one wishes Δx to determine the derivative of a function. We can determine the derivatives of many functions by applying rules of differentiation discussed below.

1. The constant function rule $\frac{dy}{dx}$

If $y = k$, then $\frac{dy}{dx} = 0$

For example, the derivative of the function $y = 5$ is 0

2. Power Function Rule

$$\text{If } y = x^r \text{ then } \frac{dy}{dx} = r x^{r-1}$$

If $y = x$ then $\frac{dy}{dx} = 1$

$$\text{If } y = x^7 \text{ then } \frac{dy}{dx} = 7x^6$$

For example if $y = x^7$ then $\frac{dy}{dx} = 7x^6$

3. Power Function Multiplied by a Constant $\frac{dy}{dx}$

If $y = Ax^r$ then $\frac{dy}{dx} = rAx^{r-1}$

For example for the function $y = 13x^3$

$$\frac{dy}{dx} = 39x^2$$

4. The Sum Rule

Suppose we have a function that is a sum of two function e.g. $H(x) = h(x) + g(x)$
 then the derivative equals the sum of the derivative of the functions. $\frac{dH}{dx} = \frac{dh}{dx} + \frac{dg}{dx}$

$$= \frac{dh}{dx} + \frac{dg}{dx} \quad \text{OR} \quad H'(x) = h'(x) + g'(x)$$

The same rule applies where a function is a sum of more than two functions.

Example

Compute the derivative of $y = 3x^2 + 5x + 7$

Solution

$$\frac{dy}{dx} = 6x + 5$$

5. The Difference Rule

The derivative of the difference of two or more functions equals the difference of the derivatives of the functions.

$$\text{If } H(x) = h(x) - g(x) : H'(x) = h'(x) - g'(x)$$

Example

Determine the derivative of $4x^2 - 2x^{-1}$

Solution

$$\frac{dy}{dx} = 8x - 2x^{-2}$$

6. Product Rule- Both Factors are Functions

The derivative of the product of two functions equals the derivative of the first function multiplied by the second function **plus** the derivative of the second function multiplied by the first function.

$$\text{If } H(x) = h(x)g(x) \text{ then } H'(x) = h'(x)g(x) + g'(x)h(x)$$

Example

Compute the derivative of $y = 3x^2(4x^3 + x^2)$

Solution $\frac{dy}{dx} = 6x(4x^3 + x^2) + 3x^2(12x^2 + 2x)$

7. Quotient Rule

The derivative of the quotient of two functions equals the derivative of the numerator times the denominator **minus** the derivative of the denominator times the numerator, all of which are divided by the square of the denominator

$$\frac{h(x)}{g(x)} \quad \frac{g(x)h'(x) - h(x)g'(x)}{[g(x)]^2}$$

$$\text{If } H(x) = \frac{h(x)}{g(x)} \text{ then } H'(x) = \frac{g(x)h'(x) - h(x)g'(x)}{[g(x)]^2}$$

Example

Find the derivative of $y = \frac{x^2}{2(x+3)}$

Solution

$$\frac{dy}{dx} = \frac{2(x+3) \cdot 2x - x^2 \cdot 2}{(2(x+3))^2} = \frac{4x(4(x+3) - 3) - 2x^2}{4(x+3)^2}$$

8. Chain Rule

The chain rule is generally applied in the determination of the derivatives of composite functions. Composite functions are functions in which another function can be considered to have taken the place of the dependent variable, thus it is always referred to as a function of a function. For example in the function,

$y = f[h(x)]$ in which $h(x)$ plays the part of the independent variable. Hence the

function can be considered as $y = f(u)$ where $u = h(x)$

This expression tells us that y is a function of u, and u is itself a function of x. in this case if y is differentiable function of u and u is a differentiable function of x then the chain rule states that

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}.$$

Example

Find the derivative of the function $y = (2x^2 + 3x)^3$

Solution

Let $y = u^3$ and $u = 2x^2 + 3x$, then

$$\frac{dy}{du} = 3u^2; \text{ and } \frac{du}{dx} = 4x + 3$$

$$\begin{aligned} \frac{dy}{dx} &= 3u^2 (4x + 3) \\ &= 3(2x^2 + 3x)^2 (4x + 3) \end{aligned}$$

For a composite function raised to a power e.g. $y = \{g(x)\}^r$ the derivative is computed as

$$\begin{aligned} \frac{dy}{dx} &= r \cdot \{g(x)\}^{r-1} \cdot g'(x) \\ &= r \cdot \{g(x)\}^{r-1} \cdot g'(x) \end{aligned}$$

Example

Find the derivative of $y = (3x^2 + 4x)^5$

Solution

$$\frac{dy}{dx} = 5(3x^2 + 4x)^4 \cdot (6x + 4)$$

7.5. ECONOMIC APPLICATION OF DERIVATIVES



a) Marginal propensity to consume

Consider a consumption function $C = 73 + 0.75Y^d$

$$MPC = \frac{dC}{dY^d} = 0.75$$

We can therefore discuss four main characteristics of the Consumption function and MPC as follows

- as the level of disposable income Y^d increases so does the corresponding level of consumption, C.
- MPC is the slope of the consumption function, and for a linear consumption function its value is constant whatever the levels of C and Y^d
- $0 < MPC < 1$; the value of the MPC is positive and less than unity
 $MPC > 0$ implies that the positive changes in Y^d give rise to positive changes in C

$MPC < 1$ implies that expenditure on consumption arising from unit additions to Y^d are less than 100%

b) Marginal propensity to save

Since savings is the portion of disposable income not consumed, the savings function associated with the consumption function given in the previous section is $S = -73 + 0.25Y^d$ and the MPS is given by the derivative

$$MPS = \frac{dS}{dY^d} = 0.25$$

Thus the characteristics of the savings function are:

- As disposable income increases so too does net savings increase
- MPS is constant whatever levels of S and disposable income, for linear savings functions
- $0 < MPS < 1$

It can be seen that there is an interrelationship between MPC and MPS.

Disposable income has two main components; consumption and savings. Thus

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$$Y^d = C + S$$

This means that any change in disposable income can be decomposed into two parts; change in consumption spending and change in savings

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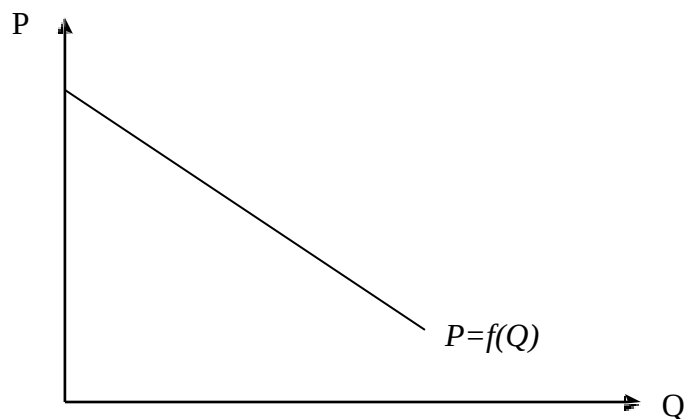
$$\Delta Y^d = \Delta C + \Delta S \rightarrow 1 = \frac{\Delta C}{\Delta Y^d} + \frac{\Delta S}{\Delta Y^d}$$

therefore

$$MPC + MPS = 1$$

c) Marginal revenue

Recall that demand for a product was defined as the mathematical relationship between the quantity of the product that is demanded and the price that the product will fetch in the market. According to the law of demand, “*other things being equal*, the quantity of a product demanded during some period of time will be larger the lower the price of that product, and vice versa”. This implies that the demand curve has a negative slope as in the figure below.



The total revenue of a firm is given by the product of price and the level of output. $TR = P.Q$. since price is a function of the quantity demanded TR can be expressed as

$$TR = P(Q).Q$$

Example

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$$\text{If } P = 3 - 0.5Q \text{ then } TR = (3 - 0.5Q)Q = 3Q - 0.5Q^2$$

Average revenue is determined by dividing the total revenue by the level of output

$$AR = \frac{TR}{Q} = \frac{P(Q) \cdot Q}{Q} = P(Q)$$

This expression suggests that the AR and demand functions are equivalent

Marginal revenue is a measure of the instantaneous rate of change of total revenue with respect to output Q.

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$$MR = \frac{d(TR)}{dQ}$$

$$\text{If } TR = 3Q - 0.5Q^2 \text{ then}$$

$$MR = 3 - Q$$

$$AR = 3 - 0.5Q$$

Exercise



The demand function for some product is $P = 16 - 0.4Q$.

- (i) Find the TR and MR functions
- (ii) Determine the value of Q for which $MR = 0$

d) Elasticity

The concept of elasticity provides a measure of proportionate change in one variable brought about by a unit change in another variable.

Consider a function $y = f(x)$

The elasticity of y with respect to x is given by

$$\frac{x}{y} \frac{dy}{dx}$$

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$$E_{y/x} = \frac{\frac{\Delta y}{y}}{\frac{\Delta x}{x}} = \frac{\Delta y}{\Delta x} \cdot \frac{x}{y}$$

In the limit $\frac{\Delta y}{\Delta x}$ reduces to $\frac{dy}{dx}$ hence

$$E_{y/x} = \frac{dy}{dx} \cdot \frac{x}{y} \text{ which constitutes point elasticity of } y \text{ with respect to } x.$$

Price elasticity of demand

Price elasticity of demand is a measure of the proportionate change in quantity demanded brought about by a corresponding proportionate change in price. It

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measure how responsive quantity demanded is to corresponding changes in price. It is measured by

$$E_{d/p} = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}} = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$$

This is the *arc elasticity of demand* with respect to price.

In the limit $\frac{\Delta Q}{\Delta P}$ reduces to $\frac{dQ}{dP}$ and the expression reduces to $E = \frac{dQ}{dP} \cdot \frac{P}{Q}$, which gives the *point elasticity of demand* with respect to price.

Since $\frac{dQ}{dP}$ is the reciprocal of the slope of the demand function, it will normally be ΔP

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negative. The price elasticity of demand will therefore normally be negative. However we customarily consider the absolute value of elasticity, rather than the negative value when making inferences

Demand can be elastic or inelastic. It is elastic if $E_{dp} < 1$ and inelastic if $E_{dp} > 1$

Elastic demand means that one percentage increase in price will cause a more than 1 percentage decrease in quantity demanded. On the other hand, inelastic demand implies that a one percentage increase in price results into a less than one percentage decrease in quantity demanded. When the percentage increase in price is exactly matched by the corresponding decrease in quantity demanded, we have a case of unitary demand. That is $E_{dp} = 1$

There is a relationship between price elasticity of demand and Total revenue. when demand is elastic, it will be accompanied by a decrease in total revenue since the increase in price leads to a more than proportionate decrease in quantity demanded. a decrease in price would lead to an increase in total revenue. for $E_{dp} < 1$,

an increase in price will give rise to a less than proportionate decrease in quantity thereby increasing total revenue. a decrease in price would cause total revenue to fall. where $E_{dp} = 1$ any change in price will result in a proportionate change in quantity

leaving total revenue fixed or constant.

Cross elasticity of demand

Cross elasticity of demand is a measure of the responsiveness of demand for one commodity to changes in the price of another commodity.

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Suppose goods 1 and 2 are related goods, and that Q_1 the quantity of good 1 demanded and

P_2 , the price of good two are related by a function $Q_1 = f(P_2)$. The

cross elasticity of good 1 with respect to price of good two would be given by

$$E_{12} = \frac{\frac{dQ_1}{dP_2} \cdot P_2}{Q_1}$$

Exercise



You are given the following supply and demand functions for a market;

Demand function : $0.04P^2 + 0.4P + 0.2Q = 20$ *supply function* : $0.15P^2 + 2P - 0.5Q = -50$

- Determine the price elasticity of demand at $P = 10$ and comment on your results
- determine the price elasticity of supply at $P = 20$ and comment on your results.

e) Marginal cost

Marginal cost is the instantaneous rate of change of total cost function with respect to output Q . alternatively it can be defined as the derivative of total cost function with respect to Q .

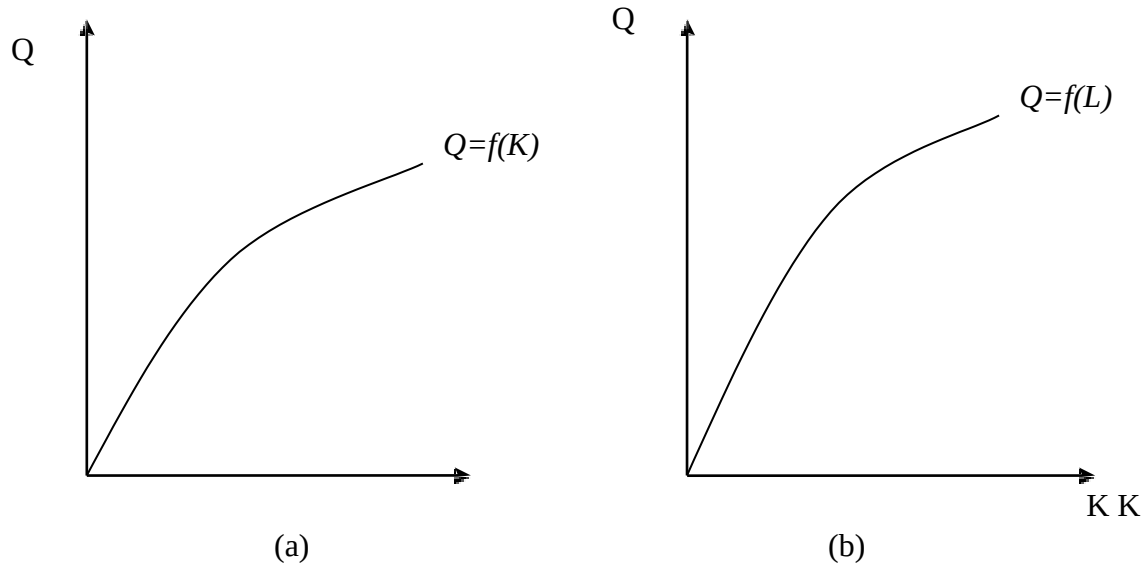
$$MC = \frac{dTC}{dQ}$$

f) Marginal product

Recall that a *production function* is a mathematical relationship representing the level of output Q which is obtained from use of one or more inputs. The general form of a production function using capital (K) and labour (L) as inputs is given as

$Q = f(K, L)$. For analytical purposes, we can assume that one of the factor inputs is held constant, and analyse the behaviour of Q with respect to the other factor input

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The figures above represent total product curves in production processes where only one factor is varied. In (a) only K is changing while in (b) only labour units are varied in the production of Q. in both cases the total product increases with increase of the relevant inputs.

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CHAPTER 8



8.0.



MAXIMA AND MINIMA



Scarcity is a basic economic problem faced by all economies. Not enough resources are available for the production of all the goods and services we may wish to consume, hence we are forced to make a choice. We have to make a decision on what commodities to produce, how much of such commodities to produce and what technique to employ in production process.

There is usually more than one technically possible alternative ways of accomplishing an economic project. With such many alternatives we have to adopt a mechanism or a criterion for choosing which alternative is most desirable. One such criterion used in economics is the objective of maximization or minimization. For example our objective may be that of maximization of revenue from taxation or it may be the minimization of costs of producing a specified level of output

Problems of maximization and minimization are collectively referred to as problems of optimization. In economics such problems include profit maximization, maximization of revenue, and minimization of costs.

The main part of a maximization or minimization problem is the *objective function*. The objective function defines the purpose or objective of the problem. The dependent variable in the objective function represents the object to be maximized or minimized. The independent variables, on the other hand, constitute the variables whose magnitudes we have to select or choose in order to maximize or minimize our objective function. For example, a business firm seeking to minimize its cost, C. The cost function given by the following function, constitute an objective function.

$$C = \frac{1}{4} Q^3 - \frac{1}{2} Q^2 + 3Q + 18$$

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C is the object of minimization. Our problem here is that of choosing the value of Q in such a way that C is minimized.

Relative and Absolute maxima and minima

The extreme value of a function can be a maximum or minimum.

A function will be considered to have reached a local maximum (also referred to as (relative maximum) at the value of $x = k$ of the independent variable if the value of the dependent variable at $x = k$ is higher than any other point in the neighborhood. similarly a function reaches a local minimum at $x = k$ if that point is lower than any other point in the neighborhood.



x x a b c d e f

The diagrams above represent graphically two functions.

In figure A, the fact that the value of the function at $x = a$ is a relative maximum does not mean that it is the highest point in the figure. The point on the curve at $x = b$ also although a relative minimum is not the lowest point on the figure. From the figure, the value of the function at $x = c$ is a global or absolute maximum and that at $x = d$ is a global or absolute minimum. These are respectively the highest and the lowest points on the function.

in figure B there is no relative maximum or minimum. the point E is the absolute maximum while the point F is the absolute minimum.

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8.1. First and second derivatives



The *first derivative* of a function is the result of differentiating the function once. For a

function $y = f(x)$ the first derivative is denoted $\frac{dy}{dx}$ or $f'(x)$

Example

Find the first derivative of the function $y = 5x^3 + 3x^2 + 7$

Solution $f'(x) =$
 $15x^2 + 6x$

The *second derivative* is the result of differentiating a function twice. $\frac{d^2y}{dx^2}$

Given a function $y = f(x)$, the second derivative is denoted $\frac{d^2y}{dx^2} = f''(x)$. the notations $\frac{d^2y}{dx^2}$

$f'(x)$ and $f''(x)$ tells us that the first and second derivatives are both functions of x ,

where as $\frac{dy}{dx}$ and $\frac{d^2 y}{dx^2}$ tells us that the differentiation has been done with respect to the variable x

Example

$$y = 5x^3 + 3x^2 + 7$$

$$\frac{dy}{dx} = 15x^2 + 6x$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \frac{dy}{dx} = 30x + 6$$

Exercise



Find the first and second derivative of the functions

a) $f(x) = x^2 + 3x$

b) $f(x) = (x^2 + 5x)^4$

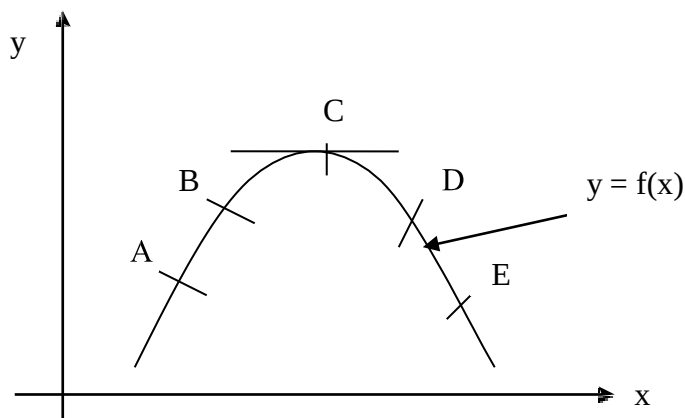
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8.2. Tests for relative maximum



The two tests commonly used to establish relative maximum and relative minimum points are the first derivative tests (First order conditions) and the second derivative test (The second order conditions) $\frac{dy}{dx}$

Consider a function $y = f(x)$ with $\frac{dy}{dx} = f'(x)$. If the function's curve is as follows;

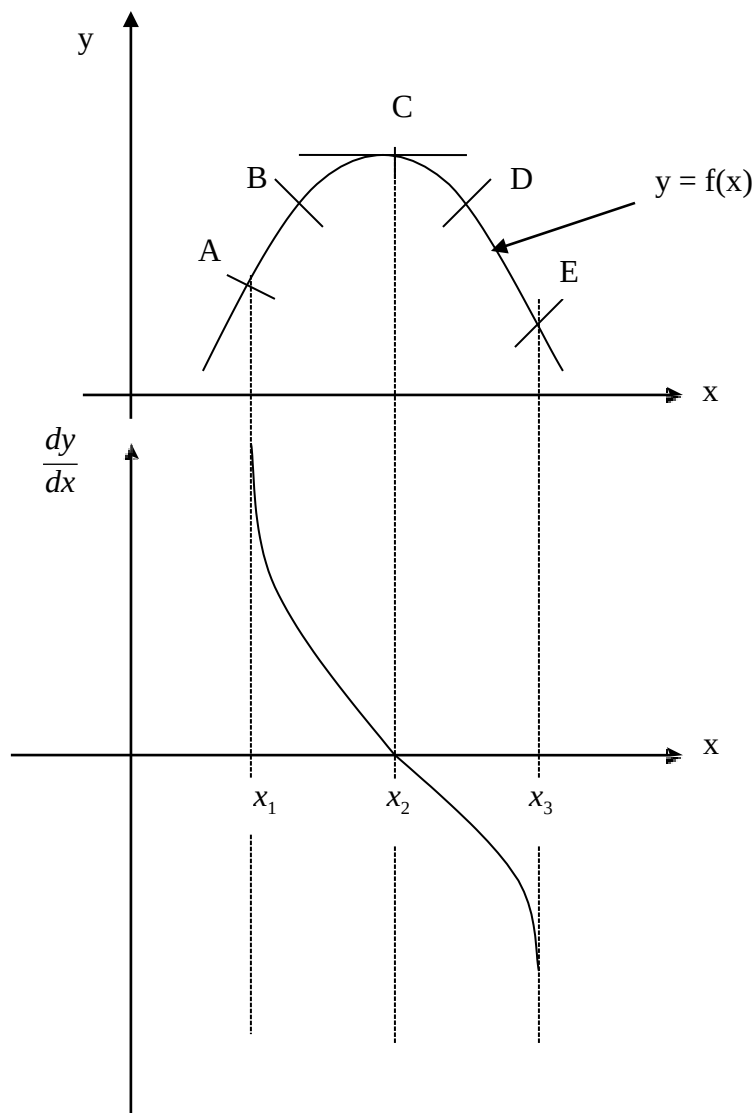


We observe that the function slopes upwards to the right between points A and C, while it slopes downwards between points C and E. this shows that the function has a positive slope between A and C but negative slope between C and E. at point C the slope is zero

Since the slope of a function is given by the derivative of that function, we can plot the graph of the derivative of the function depicted in the graph above, and study the characteristics of the resulting graph.

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The graph shows that between x_1 and x_2 the curves of $\frac{dy}{dx}$ remains above the horizontal

axis, implying that the derivative is positive. Between x_2 and x_3 the slopes of the function is negative. At point x_2 the slope is zero. From this information we derive the first and second order conditions for a relative maximum as follows.

The first order condition for a relative maximum

From the preceding figure, as y increases from A to C, the slope of the curve, and hence

derivative, remains positive. That is $\frac{dy}{dx} > 0$ for $x_1 < x < x_2$. However when y its first

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decreases from C to E the slope as well as the first derivative of the curve remains $\frac{dy}{dx}$ negative. That is $\frac{dy}{dx} \leq 0$ $x_2 \leq x \leq x_3$. At the turning point C, the slope is neither negative nor positive. It is zero. That is $\frac{dy}{dx} = 0$ $x = x_2$

Thus the first order condition for a relative maximum is given by the condition $\frac{dy}{dx} = 0$

The condition states that; *for a relative maximum, the first derivative of a function equals zero*

The second order condition for a relative maximum

We noticed that the slope of the curve changes from positive, zero to negative as one $\frac{dy}{dx}$ moves from A through C to E. this means that decreases as one moves from x_1 through x_2 to x_3

Since $\frac{dy}{dx}$ is a function of x, the fact that it is decreasing means its derivative is $\frac{d^2y}{dx^2}$ negative.

Therefore the second condition for a maximum states that: for a relative maximum the second derivative is negative. That is $\frac{d^2y}{dx^2} < 0$

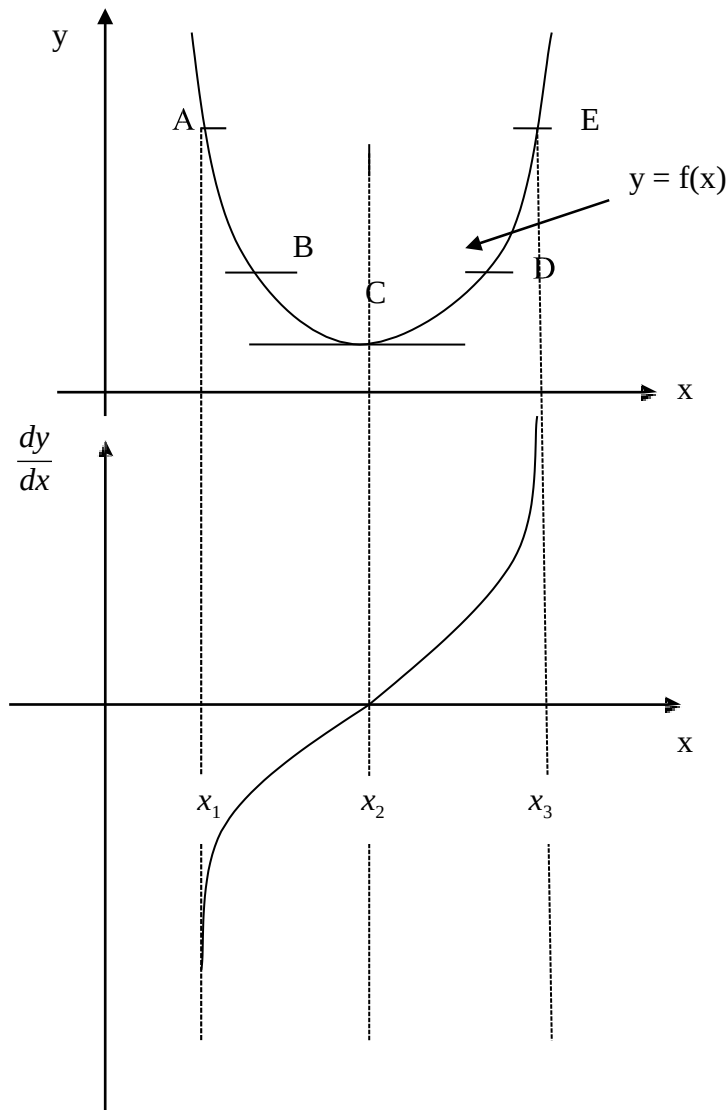
8.3 Test for relative Minimum



Just as we have done in the case of a relative maximum, we can derive the first and second conditions for a relative minimum by considering the slope of the curve at different points and the slope of the graph of its derivation of the function.

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We notice that between A and C the slope of the curve is negative, between C and E the slope is positive. At C, which is the relative minimum of the curve, the slope is zero. This implies that the first order condition for a minimum requires that the first derivative be zero and is stated; *for a relative minimum the first derivative equals zero*

however considering the graph of the first derivative, it can be seen that the slope changes from negative to zero to positive as y changes from A through to E. this implies that $f'(x)$ increases as we move from x_1 through x_2 to x_3 . The increasing value for $f'(x)$ means

that means that its derivative is increasing. Therefore the second order condition for a minimum requires that the derivative of the $f'(x)$ be positive and is stated; *for a relative minimum, the second derivative is positive.*

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Exercise



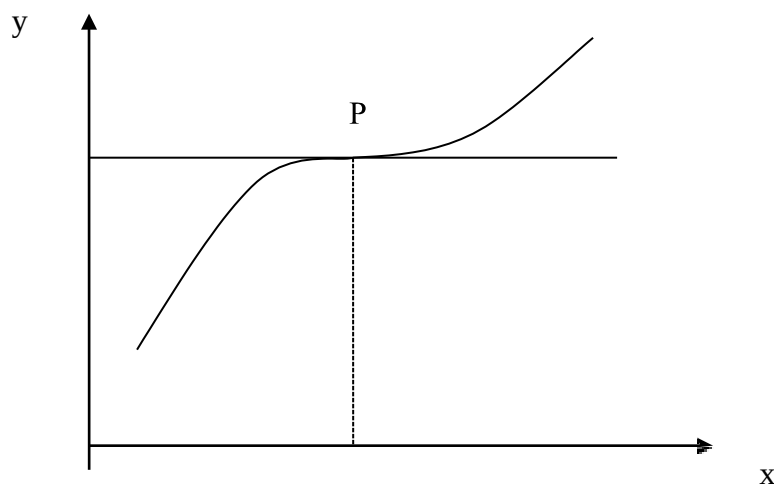
Find the critical values of following function and determine whether or not the values give rise to a local maximum or minimum

$$y = x^3 - 6x^2 + 9x + 15$$

8.4 Points of inflexion



We have defined `critical points (maximum or minimum) as a point where the first derivative of the function equals zero. However, it does not follow that all critical points result into a local maximum or minimum. Certain critical points result into what are referred to as *points of inflexion*. For example point P below



The slope at point P is zero but it is neither a local maximum nor a local minimum.

ECONOMIC APPLICATIONS OF MAXIMA AND MINIMA



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Profit maximization

The profit of a firm is given by the difference between total revenue (TR) and total costs (TC) for the firm. That is

$$\pi = TR - TC$$

But TR and TC functions can be expressed as functions of output (Q). this implies that profit itself is a function of output. Thus

$$\pi(Q) = R(Q) - C(Q)$$

to determine the profit maximizing level of output, we employ the first order condition $\frac{d\pi}{dQ}$ (F.O.C) for a maximum. That is $\frac{d\pi}{dQ} = 0$

If $\pi(Q) = R(Q) - C(Q)$ then

$$\frac{d\pi}{dQ}$$

$$= R'(Q) - C'(Q) = 0 \rightarrow R'(Q) = C'(Q)$$

Since $R'(Q)$ the derivative of the total revenue is marginal revenue, and $C'(Q)$ the derivative of total cost is the marginal cost, it implies that the profit is maximized when the condition $MR = MC$ is met. The quantity Q that satisfies this condition is referred to as the profit maximizing output. it is also referred to as $\bar{Q}$ the optimal output.

The second order condition for profit maximization requires that $\frac{d^2\pi}{dQ^2} < 0$

$$\frac{d^2\pi}{dQ^2}$$

$$< 0$$

$$\frac{d^2\pi}{dQ^2}$$

$$\frac{d}{dQ} \left(\frac{d\pi}{dQ} \right) = \frac{d}{dQ} (MR - MC) = \frac{d(MR)}{dQ} - \frac{d(MC)}{dQ} < 0 \rightarrow \frac{d(MR)}{dQ} < \frac{d(MC)}{dQ}$$

The expression implies that for maximum profit the first derivative (slope) of the marginal revenue function must be less than the first derivative (slope) of the marginal cost function

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Example

Suppose the AR and AC of a firm are defined by the following functions

$$AR = 4 - \frac{1}{4}Q$$

$$AC = \frac{4}{Q} + 2 - 0.3Q + 0.05Q^2$$

We can determine the level of q and p that will maximize profits of the firm.

From the AR and AC functions we can obtain TR and TC curves

$$TR = AR \cdot Q = 4Q - \frac{1}{4}Q^2$$

$$TC = AC \cdot Q = 4 + 2Q - 0.3Q^2 + 0.05Q^3$$

Profit which is TR-TC can be computed

$$\begin{aligned}\Pi &= 4Q - 0.25Q^2 - 4 - 2Q + 0.3Q^2 - 0.05Q^3 \\ &= 2Q - 0.05Q^2 - 0.05Q^3 - 4\end{aligned}$$

$$\text{By F.O.C } \frac{d\Pi}{dQ} = 2 - 0.1Q -$$

$$0.15Q^2 = 0$$

solving for Q yields

$$Q = -\frac{1}{3} \text{ or } 4$$

Since negative output makes no meaning, the critical point occurs at Q = 4 By

second order condition

$$\frac{d\pi}{dQ} = 2 - 0.1Q - 0.15Q^2$$

dQ

$$\frac{d^2\pi}{dQ^2} = -0.1$$

$$-0.3Q \leq 0 \quad dQ^2$$

taking the value $Q = 4$

$\frac{d\pi}{dQ}$

$$= -0.1 - 0.3 \times 4 = -0.1 - 1.2 = -1.3 \leq 0$$

this satisfies the condition for a maximum.

Maximization of tax revenue

Suppose the government intends to raise tax by imposing a per unit tax on products of firms. Suppose also that the total tax T collected by government is a function of output given by $T = tQ$ where t is the tax rate.

Consider a firm with TC and TR functions given as below

$$TC = \pi_0 + \pi_1 Q + \pi_2 Q^2$$

$$TR = \pi_1 Q - \pi_2 Q^2$$

The imposition of the tax raises the total cost of the firm by the amount of tax so that its after tax total cost function becomes

$$TC_t = \pi_0 + \pi_1 Q + \pi_2 Q^2 + tQ$$

the profits of the firm will therefore be lowered by the amount of tax. the profits after tax is given by

$$\pi_t = (\pi_1 Q - \pi_2 Q^2) - [\pi_0 + \pi_1 Q + \pi_2 Q^2 + tQ]$$

If the government objective is to maximize tax revenue, then it will choose some tax rate t for which tax revenue $T = tQ$ is maximized. Where $\bar{Q}$ is the optimal or profit

maximizing output for the firm. The function $T = tQ$ is objective function for the

government, where T , tax revenue, is the object and t is the variable whose value it wants to choose in such a manner as to maximize T . To determine $\bar{Q}$ we determine Q that maximizes $\pi_t = (\pi_1 Q - \pi_2 Q^2) - [\pi_0 + \pi_1 Q + \pi_2 Q^2 + tQ]$ from first order conditions

$$\frac{dQ}{dt} = \alpha_1 - 2\alpha_2 Q - \beta_1 - 2\beta_2 Q - t = 0$$

$$-2(\alpha_2 - \beta_2)Q = -\alpha_1 + \beta_1 + t$$

$$Q = \frac{(\alpha_1 - \beta_1 - t)}{2(\alpha_2 - \beta_2)}$$

This implies that the tax revenue function for the government becomes

$$T = tQ = \frac{t(\alpha_1 - \beta_1 - t)}{2(\alpha_2 - \beta_2)}$$

)

In order to maximize T the first order condition requires that $T'(t) = 0$. Therefore

$$\frac{dT}{dt} = \frac{\alpha_1 - \beta_1 - 2t}{2(\alpha_2 - \beta_2)} = 0 \rightarrow \alpha_1 - \beta_1 - 2t = 0$$

The second condition

$$\frac{d^2T}{dt^2} = \frac{-2}{2(\alpha_2 - \beta_2)} = \frac{-1}{\alpha_2 - \beta_2} < 0$$

hence dt

$$t = \frac{\alpha_1 - \beta_1}{2} \text{ gives the optimum tax rate.}$$

Maximization of revenue

Consider the demand facing a firm given by the following linear function

$$P = \alpha - \beta Q$$

The total revenue function for the firm is therefore

$$TR = Q.P = Q(a - bQ) = aQ + bQ^2$$

The equilibrium output $\bar{Q}$ level is obtained by first order conditions.

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$$TR = Q.P = Q(a - bQ) = aQ + bQ^2$$

$$\frac{dTR}{dQ} = a - 2bQ = 0 \rightarrow Q = \frac{a}{2b}$$

$$Q = \frac{a}{2b}$$

To confirm that the equilibrium output is the revenue maximizing output level, we check the 2nd order condition

$$\frac{d^2TR}{dQ^2} = -2b < 0$$

Minimization of cost

Minimization of minimum cost

Consider a cubic total cost function

$$TC = c_0 + c_1Q + c_2Q^2 + c_3Q^3$$

We can obtain the MC function $\frac{dTC}{dQ}$

$$MC = \frac{dTC}{dQ} = c_1 + 2c_2Q + 3c_3Q^2$$

The output $\bar{Q}$ that constitute the critical level for the MC function will satisfy the first order condition $\frac{dMC}{dQ} = 0$

$$\frac{dQ}{dQ} = 0 = -2Q^2 + 6Q_3Q$$

$$Q = \frac{Q_2}{3Q_3}$$

The second derivative of MC function is d^2MC

$$\frac{d^2MC}{dQ} = 6Q_2 - 3Q_3$$

$$Q = Q_2$$

which satisfies the second order condition for minimum. Therefore Q_2 is the output level that minimizes MC.

Minimum average cost

Consider average cost function

$$AC = \frac{Q_0}{Q} + Q_1 + Q_2 Q$$

The average cost function will attain its minimum when its first derivative with respect to Q is equal to zero $\frac{dAC}{dQ} = -\frac{Q_0}{Q^2} + Q_2 = 0 \rightarrow Q_2 =$

$$\frac{Q_0}{Q^2} \rightarrow \bar{Q} = \sqrt{\frac{Q_0}{Q_2}}$$

but negative values are meaningless in economics, hence we discard the negative value and take the positive value for the critical value that minimizes average costs.

TOPIC 9



9.0 PARTIAL DIFFERENTIATION

Your text here 3



Introduction



We have already discussed and given examples of functions in economic analysis of one independent variable (the univariate function). Although in many cases this type of representation gives a fairly accurate approximation of reality, there are several cases in which it is extremely inadequate. In such cases it is necessary to extend the concept of the function from univariate to multivariate form. For example for a function $f(x)$ we can have

$$y = \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \alpha_3 x^4 \text{ or simply as } y = f(x_1, x_2, x_3, x_4)$$

The concept of derivatives has been defined for univariate functions. In order to differentiate the multivariate function we need to extend the concept of derivatives to the multivariate case.

In multivariate functions, the derivative gives the instantaneous rate of change of the function with respect to one of the independent variables in the function. Since the multivariate function has two or more independent variables, the derivative of the

function is considered separately for each one of the independent variables. This requires that while calculating the derivative of the function with respect to one of its independent variables, the remaining independent variables are assumed to be constant.

The term *partial derivative* is used to denote the derivative of a multivariate function. The derivative is known as partial because the process of differentiation is done with respect to one independent variable at a time. Partial differentiation therefore refers to the process of determining the partial derivatives of a function.

The symbol

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0

$\frac{\partial y}{\partial x}$ denotes the partial derivative of y with respect to x . We can

$\frac{\partial y}{\partial x_1}$

1

alternatively use f_{x_1} or f_1 for the same purposes. The partial derivative of y with respect

to x_2 is represented by the symbol $\frac{\partial y}{\partial x_2}$ or f_{x_2}

Examples

Consider a function $y = 5x_1 + 7x_2$

When differentiating partially with respect to x_1 we assume x_2 to be constant. Similarly,

when differentiating with respect to x_2 we assume x_1 to be constant. Therefore

$$\frac{\partial y}{\partial x_1} = 5 \text{ while } \frac{\partial y}{\partial x_2} = 7$$

If the partial derivatives of a function are functions of the can be re-differentiated partially so as to yield higher order partial derivatives.

Find $\frac{\partial y}{\partial x_1}$ and $\frac{\partial y}{\partial x_2}$ for the function $y = 2x_1 + 3x_2$

$$\frac{\partial y}{\partial x_1} \text{ and } \frac{\partial y}{\partial x_2}$$

$$\frac{\partial y}{\partial x_1} = 2 \text{ and } \frac{\partial y}{\partial x_2} = 3$$

Exercise



Given $y = x_1^2 + 2x_2^2$

$\frac{\partial y}{\partial x_1}$ and $\frac{\partial y}{\partial x_2}$

Find

$\frac{\partial y}{\partial x_1}$ and $\frac{\partial y}{\partial x_2}$

Rules for partial differentiation



1. Sum and difference rule

The partial derivative of a sum or difference of two functions $f(x_1, x_2)$ and $g(x_1, x_2)$ with respect to x_1 is the sum or difference of the partial derivative of the two functions

with respect to x_1 . That is;

if $y = f(x_1, x_2) \pm g(x_1, x_2)$ then

$$\frac{\partial y}{\partial x_1} = \frac{\partial f}{\partial x_1}(x_1, x_2) \pm \frac{\partial g}{\partial x_1}(x_1, x_2) \text{ and } \frac{\partial y}{\partial x_2} = \frac{\partial f}{\partial x_2}(x_1, x_2) \pm \frac{\partial g}{\partial x_2}(x_1, x_2)$$

Example

Consider a function $y = 13x_1^2 x_2 + 21x_1 x_2^3$

$$\frac{\partial y}{\partial x_1} = 26x_1 + 21x_2$$

$$\frac{\partial y}{\partial x_1}$$

$$\frac{\partial y}{\partial x_2} = 13x_1 + 63x_2$$

2. Product rule

The partial derivative of the product of two functions $f(x_1, x_2)$ and $g(x_1, x_2)$ with respect to x_1 equal to the partial derivative of the first function with respect to x_1

times the second function PLUS the partial derivative of the second function times the first function with respect to x_1 . That is If $y = f(x_1, x_2).g(x_1, x_2)$ then

$$\frac{\partial y}{\partial x_1} = f_1 g + g_1 f \quad \text{and} \quad \frac{\partial y}{\partial x_2} = f_2 g + g_2 f$$

$$\frac{\partial y}{\partial x_1}$$

Example

Suppose $y = (3x_1^2 + x_2)(x_1 + x_2^2)$ then

$$\frac{\partial y}{\partial x_1} = 6x_1(x_1 + x_2^2) + 1(3x_1^2 + x_2)$$

$$\frac{\partial y}{\partial x_1}$$

$$\frac{\partial y}{\partial x_2} = 2x_1 + 2x_2(3x_1^2 + x_2)$$

$$\frac{\partial y}{\partial x_2}$$

3. Quotient rule

The partial derivative of a quotient $y = \frac{f(x_1, x_2)}{g(x_1, x_2)}$ with respect to x_1 equal to the partial derivative of the numerator with respect to x_1 times the denominator MINUS the

derivative of the denominator with respect to x_1 times the numerator MINUS the

partial derivative of the denominator with respect to x_1 times the numerator all divided by the square of the denominator. That is

$$\frac{\partial y}{\partial x_1} = \frac{f_1 g - g_1 f}{g^2}$$

Example

If $y = \frac{x_1 x_2 + x_2^2}{x_1 x_2}$ then

$$\frac{\partial y}{\partial x_1} = \frac{f_1 g - g_1 f}{g^2} = \frac{2x_2(x_1 x_2) - x_1 x_2^2}{(x_1 x_2)^2} = \frac{2x_2 - x_2}{x_1 x_2} = \frac{x_2}{x_1 x_2} = \frac{1}{x_1}$$

4. Partial derivative of a function of two independent variables raised to a power r

Suppose $y = [f(x_1, x_2)]^r$ then the derivative with respect to x_1 is given by the

r times the function $f(x_1, x_2)$ raised to power $(r-1)$ all multiplied by the partial derivative of the function with respect to x_1 . That is

$$\frac{\partial y}{\partial x_1} = r[f(x_1, x_2)]^{r-1} \cdot f_1$$

Example

If $y = [3x_1^2 + x_2^2]^5$ Then

$$\frac{\partial y}{\partial x_1} = 5 [3x_1^2 + x_2^2]^4 \cdot 3x_1$$

$$\frac{\partial y}{\partial x_2} = 5 [3x_1^2 + x_2^2]^4 \cdot (2x_2)$$

ECONOMIC APPLICATION OF PARTIAL DERIVATIVES

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